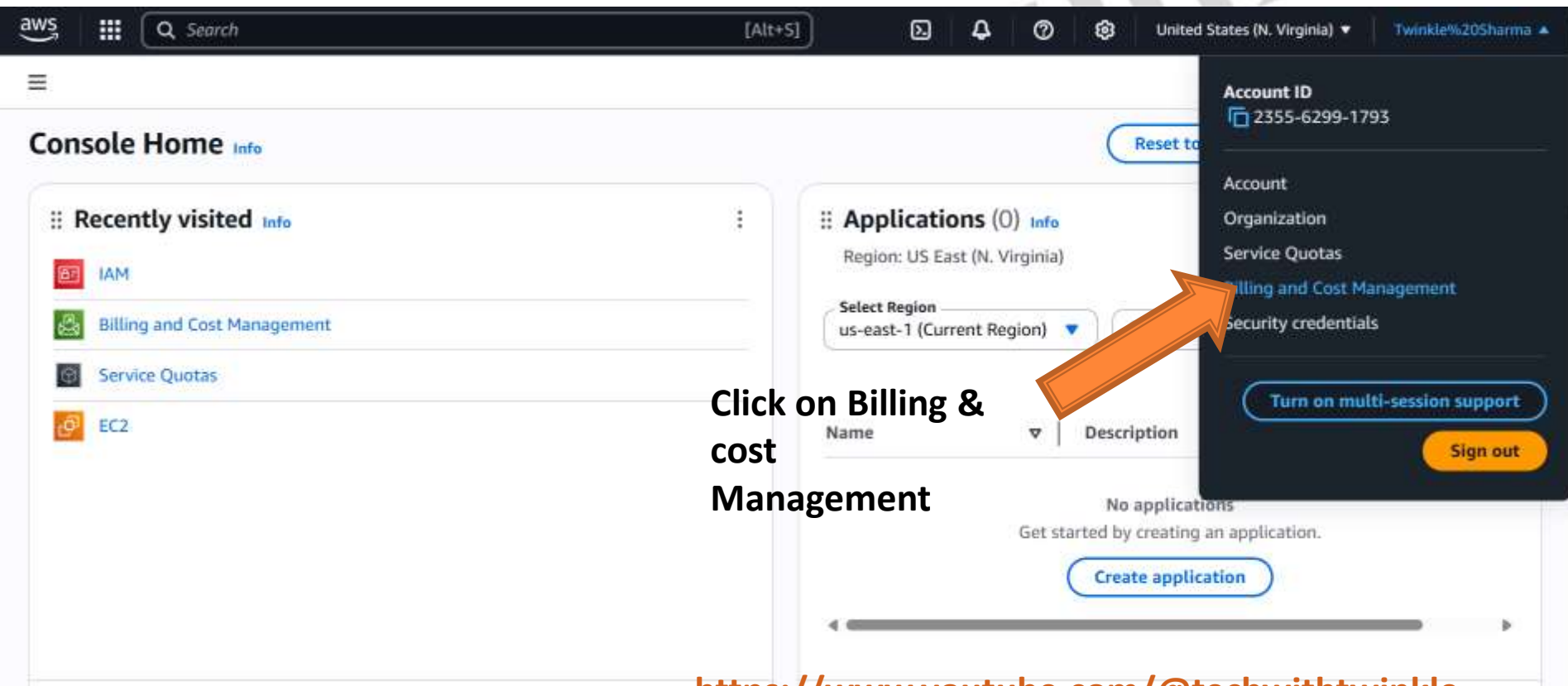


EC2 Fundamentals

SAA-C03

AWS Budget Setup

- ▶ Login your root user or IAM Account



<https://www.youtube.com/@techwithtwinkle>

aws [Alt+S] Global tv

Billing and Cost Management home

Provide feedback Need help? Ask Q Reset layout

Cost summary

Month-to-date cost
Data unavailable

Last month's cost for same time period
Data unavailable

Total forecasted cost for current month
Access denied

Last month's total cost
Data unavailable

Cost monitor

Budgets status
Setup required

No budget created

Cost anomalies status (MTD)
None detected

1 monitor(s) active

Cost breakdown

Recommended actions (1)

Choose billing view New
Primary view

Home
Getting Started
Billing and Payments
Bills
Payments
Credits
Purchase Orders
Cost and Usage Analysis
Cost Explorer

Here you will see bill & payments

Here you will see cost details, As this is new account so no summary available

aws [Alt+S] Global

Billing and Cost Management home

Provide feedback Need help? Ask Q Reset layout

Cost Organization

Cost Categories
Cost Allocation Tags
Billing Conductor

Budgets and Planning

Budgets New
Budgets Reports
Pricing Calculator New

Savings and Commitments

Cost Optimization Hub New

Savings Plans

Overview
Inventory
Recommendations
Purchase Analyzer New

Cost summary

Month-to-date cost
Data unavailable

Last month's cost for same time period
Data unavailable

Total forecasted cost for current month
Access denied

Last month's total cost
Data unavailable

Cost monitor

Budgets status
Setup required

No budget created

Cost anomalies status (MTD)
None detected

1 monitor(s) active

View bill

Cost breakdown

Recommended actions (1)

Click on Budget

<https://www.youtube.com/@techwithtwinkle>

aws [Search] [Alt+S] Global Twinkle%20Sharma

Billing and Cost Management > Budgets

Billing and Cost Management

Choose billing view [New](#)

Primary view

Home
Getting Started
Billing and Payments
Bills
Payments
Credits
Purchase Orders
Cost and Usage Analysis
Cost Explorer

AWS Budgets

Set custom budgets that alert you when you exceed your budgeted thresholds

AWS Budgets is your hub for creating, tracking, and inspecting your budgets.

Start tracking your AWS costs and usage

Once you have a budget created, AWS Budgets allows you to create budgets, forecast spend, and take action on your costs and usage from a single location.

[create a budget](#)

Pricing (US)

There is no additional charge for using AWS Budgets. You only pay for configured actions that go beyond the

How it works

Click on Create Budget

aws [Search] [Alt+S] Global Twinkle%20Sharma

Billing and Cost Management > Budgets > Create budget

Billing and Cost Management

Choose billing view [New](#)

Primary view

Home
Getting Started
Billing and Payments
Bills
Payments
Credits
Purchase Orders
Cost and Usage Analysis
Cost Explorer

For now – I am using template

Selecting Zero spend budget

☒ **Use a template (simplified)**
Use the recommended configurations. You can change some configuration options after the budget is created.

☐ **Customize (advanced)**
Customize a budget to set parameters specific to your use case. You can customize the time period, the start month, and specific accounts.

Templates - new

Choose a template that best matches your use case.

☒ **Zero spend budget**
Create a budget that notifies you once your spending exceeds \$0.01.

☐ **Monthly cost budget**
Create a monthly budget that notifies you if you exceed, or are forecasted to exceed, the budget amount.

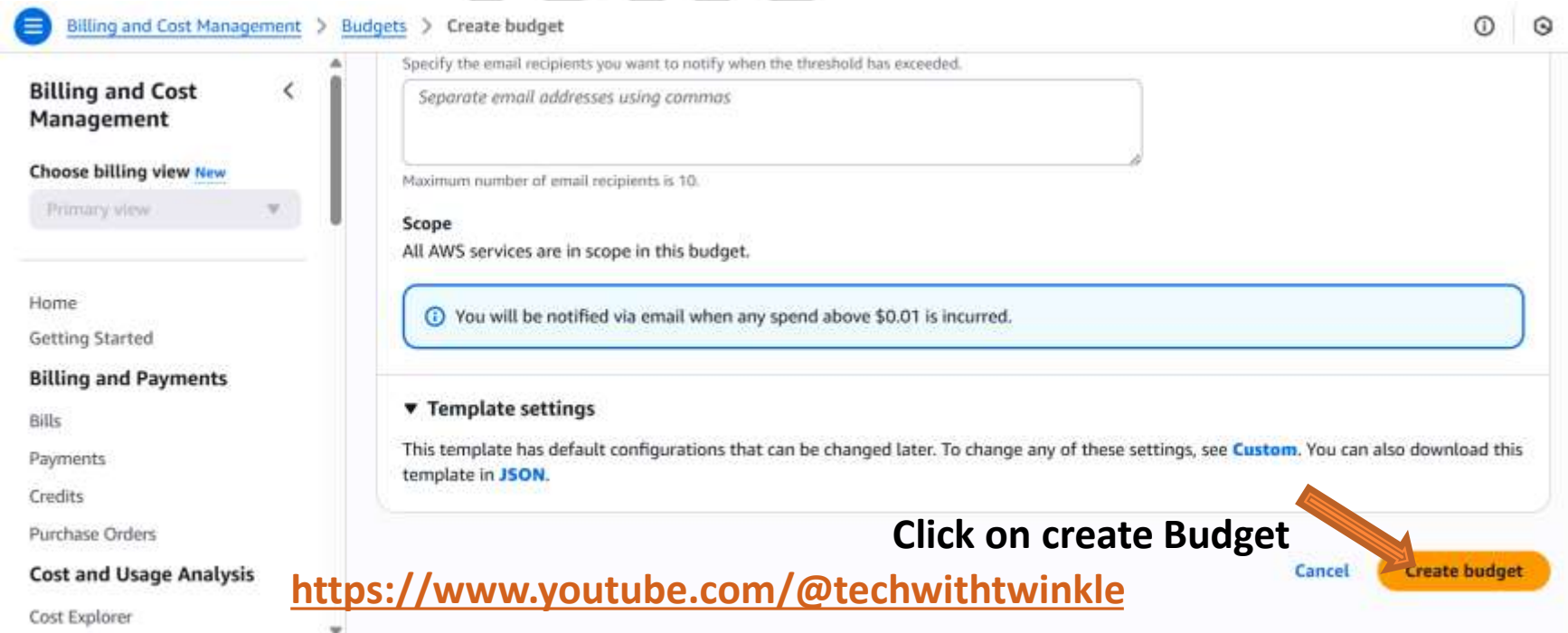
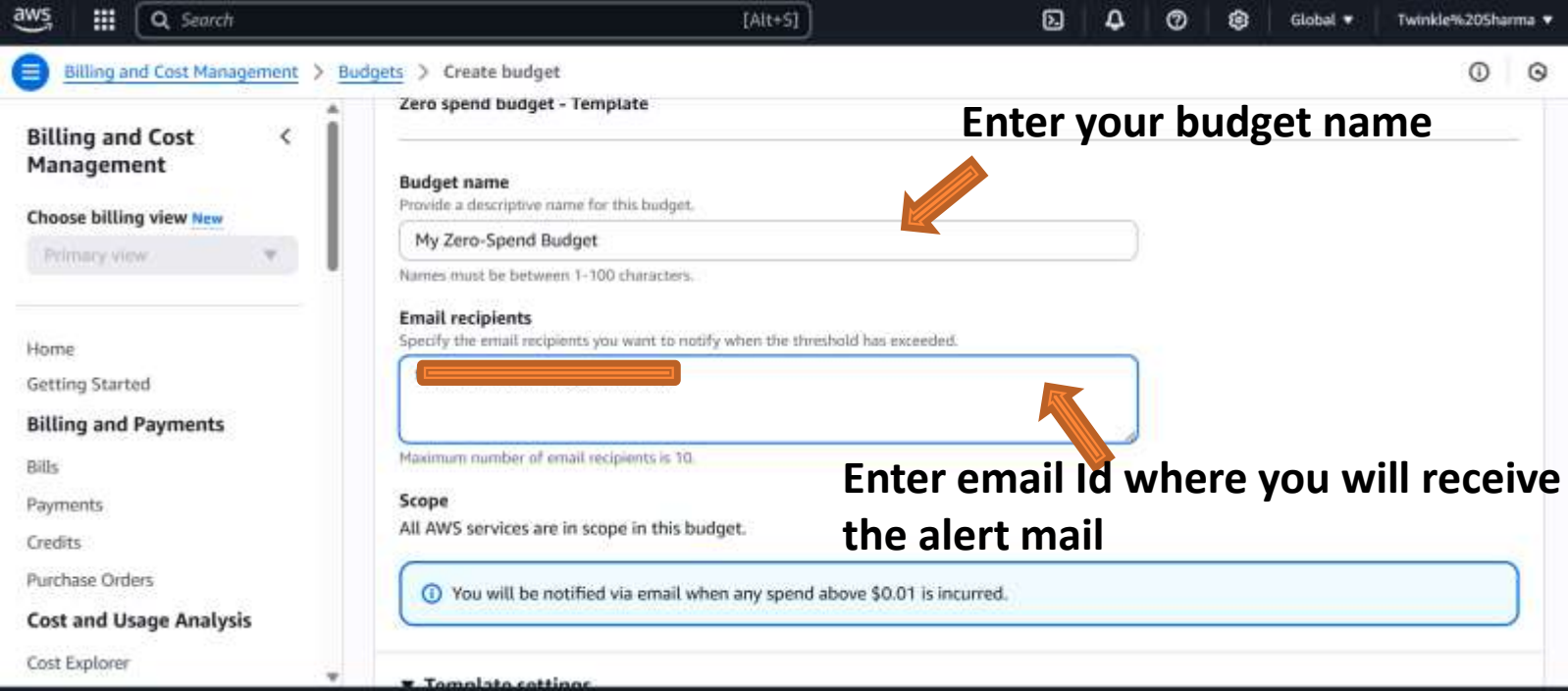
☐ **Daily Savings Plans coverage budget**
Create a coverage budget for your Savings Plans that notifies you when you fall below the defined target.

☐ **Daily reservation utilization budget**
Create a utilization budget for your reservations that notifies you when you fall below the defined target.

Zero spend budget - Template

Budget name

<https://www.youtube.com/@techwithtwinkle>



Billing and Cost Management > Budgets > Overview

Billing and Cost Management

Choose billing view [New](#)

Primary view

Home
Getting Started
Billing and Payments
Bills
Payments
Credits
Purchase Orders
Cost and Usage Analysis
Cost Explorer

Budgets (1) Info

[Download CSV](#) [Actions](#) [Create budget](#)

Find a budget Type - Show all budgets < 1 > ⚙️

☐	Name	Thresholds	Budget	Amount ...	Forecast...	Current vs. budget
☐	My Zero-Spend Budget	OK	\$1.00	\$0.00	-	

Our Budget is created

Same way you can create Monthly budget – But in that you have to set budget amount shown below

aws Search [Alt+S] Global Twinkle%20Sharma

Billing and Cost Management > Budgets > Create budget

Billing and Cost Management

Choose billing view [New](#)

Primary view

Home
Getting Started
Billing and Payments
Bills
Payments
Credits
Purchase Orders
Cost and Usage Analysis
Cost Explorer

Monthly cost budget - Template

Budget name
Provide a descriptive name for this budget.

My Monthly Cost Budget

Names must be between 1-100 characters.

Enter your budgeted amount (\$)
Last month's cost: \$0.00

100

Email recipients
Specify the email recipients you want to notify when the threshold has exceeded.

Separate email addresses using commas

Maximum number of email recipients is 10.

Scope
All AWS services are in scope in this budget.

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Amazon EC2

- ▶ **Amazon EC2 (Elastic Compute Cloud)** is a core AWS service that provides **scalable virtual servers in the cloud**, allowing you to run applications with full control over:
 - ▶ OS (Linux/Windows)
 - ▶ Network
 - ▶ Storage
 - ▶ Compute resources

<https://www.youtube.com/@techwithtwinkle>

Concept	Description
Instance	A virtual machine you launch in AWS
AMI	Amazon Machine Image (OS template)
EBS Volume	Elastic Block Storage attached to EC2
Key Pair	Secure login via SSH (Linux) or RDP (Windows)
Security Group	Acts like a virtual firewall
Elastic IP	A static public IP address

EC2 Instance Types

EC2 offers **multiple families** of instances optimized for different workloads:

Family	Optimized For	Examples
<code>t</code> (General)	General purpose	<code>t3.micro</code> , <code>t4g.small</code>
<code>m</code> (Balanced)	Compute + memory balance	<code>m5.large</code> , <code>m6a.large</code>
<code>c</code> (Compute)	High compute performance	<code>c7g.large</code> , <code>c6i.xlarge</code>
<code>r</code> (Memory)	RAM-intensive applications	<code>r6g.large</code> , <code>r5b.2xlarge</code>
<code>i</code> , <code>d</code> (Storage)	High IOPS/local storage	<code>i4i.large</code> , <code>d3.2xlarge</code>
<code>g</code> , <code>p</code> (GPU)	Machine learning, graphics	<code>g5.xlarge</code> , <code>p4d.24xlarge</code>
<code>h</code> , <code>z</code> , <code>u</code>	Specialized needs	High memory, high GHz CPUs

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Instance Sizing & Configuration

▶ How to Choose Instance Size?

Factor	What to Consider
vCPUs	Amount of processing power
RAM (GiB)	Memory needed for app/database
Network Bandwidth	Data transfer per second (Gbps)
Storage	Use EBS for persistence, instance store for temp data
Workload type	Web server, ML, DB, gaming, analytics?
Region Availability	Not all instance types are in all regions

Example: A web server may run on `t3.micro` , but a high-performance database might need `r6g.4xlarge` .

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EC2 Configuration Checklist

Element	Purpose	Example
AMI	OS Template	Amazon Linux, Ubuntu, Windows
Instance Type	Size/performance	<code>t2.micro</code> , <code>c6g.large</code>
Key Pair	Secure SSH or RDP access	<code>ec2-keypair.pem</code>
Network (VPC/Subnet)	Determines IP and route settings	VPC with public subnet
Security Group	Controls inbound/outbound rules	Allow port 22 for SSH
User Data	Bootstrap instance at launch	Install Apache, update system
Storage (EBS)	Persistent storage	30 GiB gp3 root volume
IAM Role	Grants access to S3, CloudWatch, etc.	<code>EC2S3AccessRole</code>
Tags	Key-value metadata	<code>Environment=Dev</code>

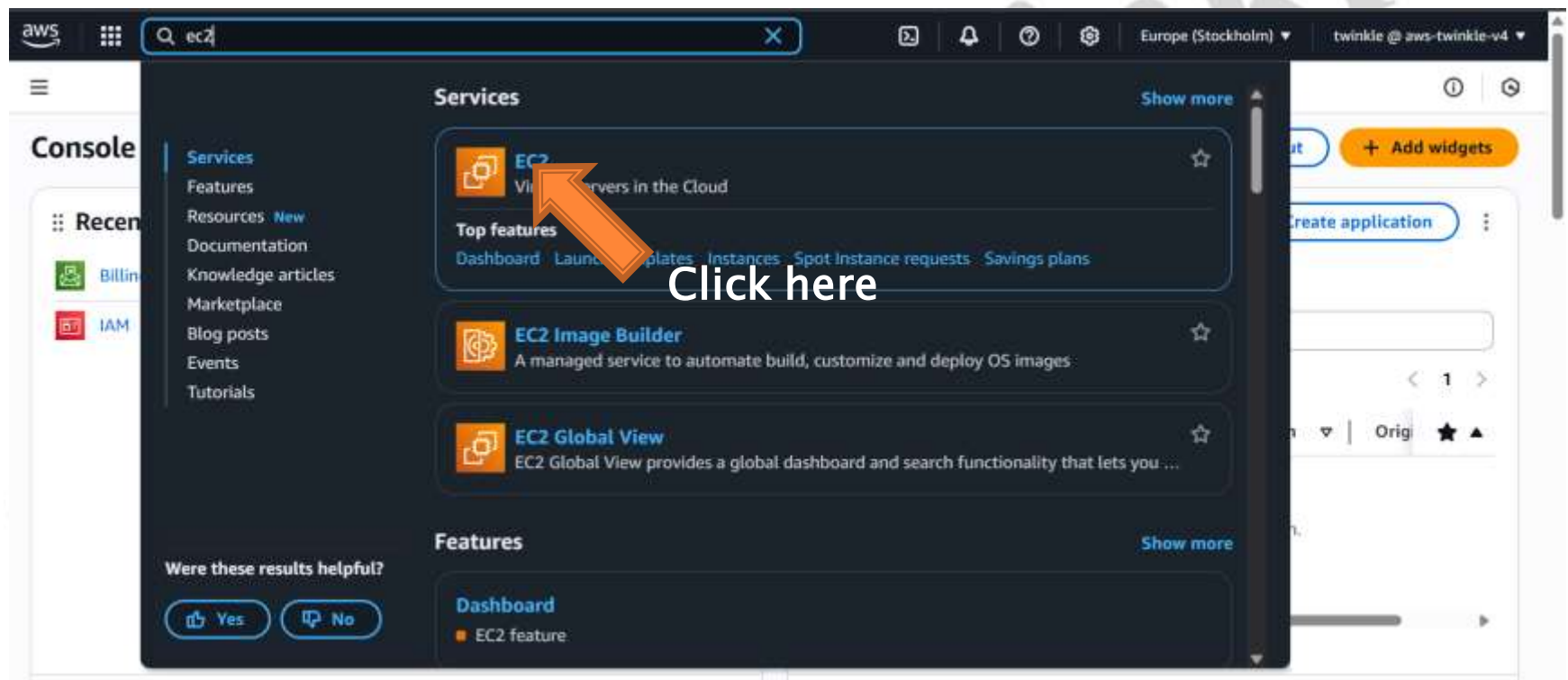
<https://www.youtube.com/@techwithtwinkle>

Best Practices

- ▶ Use IAM roles instead of access keys
- ▶ Restrict ports in Security Groups
- ▶ Use user data for consistent setup
- ▶ Use Auto Scaling for high availability
- ▶ Monitor with CloudWatch
- ▶ Take regular snapshots of EBS
- ▶ Use lifecycle policies to stop unused instances

Create EC2 instance with EC2 User Data to have a Website Hands On

Login to your IAM Account than Search EC2



<https://www.youtube.com/@techwithtwinkle>

Before creating Instance select nearby location of your location

Click Here

Launch an instance

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags

Name: My First Server

Application and OS Images (Amazon Machine Image)

Search our full catalog including 1000s of application and OS images

Summary

Number of instances: 1

Software Image (AMI): Amazon Linux 2023 AMI 2023.8.2...

Virtual server type (instance type): t3.micro

Firewall (security group): New security group

Launch instance

aws Search [Alt+S]

EC2 > Instances > Launch an instance

Application and OS Images (Amazon Machine Image) [Info](#)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below

Search our full catalog including 1000s of application and OS images

Quick Start

Amazon Linux
aws

Ubuntu
ubuntu

Windows
Microsoft

Red Hat
Red Hat

SUSE Linux
SUSE

Debian
debian

[Browse more AMIs](#)
Including AMIs from AWS, Marketplace and the Community

Free tier eligible

Summary

Number of instances [Info](#)

1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.8.2...[read more](#)
ami-07f0dc6c4fe353ff1

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

[Cancel](#) [Launch instance](#) [Preview code](#)

Select Amazon Linux

aws Search [Alt+S]

EC2 > Instances > Launch an instance

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI
ami-07f0dc6c4fe353ff1 (64-bit (x86), uefi-preferred) / ami-03dc3e3befcd41326 (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

Description

Amazon Linux 2023 (kernel-6.1) is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Amazon Linux 2023 AMI 2023.8.20250707.0 x86_64 HVM kernel-6.1

Architecture	Boot mode	AMI ID	Publish Date	Username
64-bit (...)	uefi-preferred	ami-07f0dc6c4fe353ff1	2025-07-08	ec2-user

Verified provider

Summary

Number of instances [Info](#)

1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.8.2...[read more](#)
ami-07f0dc6c4fe353ff1

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

[Cancel](#) [Launch instance](#) [Preview code](#)

For Now, I am selecting free tier one you can select as per your requirement

<https://www.youtube.com/@techwithtwinkle>

By clicking here you can compare instance types

For Now, I am selecting free tier one you can select as per your requirement

Instance type Info | Get advice

Instance type

t3.micro

Family: t3 2 vCPU 1 GiB Memory Current generation: true

On-Demand RHEL base pricing: 0.04 USD per Hour

On-Demand Windows base pricing: 0.0204 USD per Hour

On-Demand Ubuntu Pro base pricing: 0.0147 USD per Hour

On-Demand Linux base pricing: 0.0112 USD per Hour

On-Demand SUSE base pricing: 0.0112 USD per Hour

Free tier eligible

All generations

Compare instance types

Additional costs apply for AMIs with pre-installed software

Key pair (login) Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

Summary

Number of instances Info

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.8.2...read more

ami-07f0dc6c4fe353ff1

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Launch Instance

Preview code

Click Here to create key pair

Key pair (login) Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

Select

Create new key pair

Network settings Info

Network Info

vpc-04c59650b26f4b576

Subnet Info

No preference (Default subnet in any availability zone)

Auto-assign public IP Info

Enable

Summary

Number of instances Info

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.8.2...read more

ami-07f0dc6c4fe353ff1

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Cancel

Launch Instance

Preview code

<https://www.youtube.com/@techwithtwinkle>

aws [Search] [Alt+S] Asia Pacific (Hyderabad) twinkl@aws-twinkle-v4

EC2 > Instances > Launch an instance

Key pair (login)

You can use a key pair to securely connect to your instance.

Key pair name - required

Select

Network

Network vpc-04c59690/36f4b1576

Subnet No preference (Default subnet in any available region)

Auto-assign public IP Enable

Create key pair

Key pair name
Key pairs allow you to connect to your instance securely.

firstkeypair

The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type

☒ RSA
RSA encrypted private and public key pair

☐ ED25519
ED25519 encrypted private and public key pair

Private key file format

☒ .pem
For use with OpenSSH

☐ .ppk
For use with PuTTY

Cancel Create key pair

Enter Key pair name

Select key pair type

After all this selection, click on create key pair

Select private key file format

aws [Search] [Alt+S] Asia Pacific (Hyderabad) twinkl@aws-twinkle-v4

EC2 > Instances > Launch an instance

Auto-assign public IP Enable

Additional charges apply when outside of free tier allowance

Firewall (security groups)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group ☐ Select existing security group

We'll create a new security group called 'launch-wizard-1' with the following rules:

☒ Allow SSH traffic from Anywhere 0.0.0.0/0

☐ Allow HTTPS traffic from the internet

☐ Allow HTTP traffic from the internet

Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Summary

Number of instances 1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.8.2...read more
ami-07f0dc6c4fe353ff1

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Cancel Launch instance Preview code

<https://www.youtube.com/@techwithtwinkle>

Select this

aws Search [Alt+S]

EC2 > Instances > Launch an instance

Configure storage

1x 8 GiB gp3 Root volume, 3000 IOPS, Not encrypted

Free tier eligible customers can get up to 30 GB of EBS General Purpose (SSD) or Magnetic storage

Add new volume

Click refresh to view backup information
The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.

0 x File systems

Advanced details

Summary

Number of Instances 1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.8.2...read more
ami-07f0dc6c4fe353ff1

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Cancel Launch instance Preview code

Click on advanced details

aws Search [Alt+S]

EC2 > Instances > Launch an instance

User data - optional

Upload a file with your user data or enter it in the field.

Choose file

Open the Code Folder in your notes files
Go to EC2 Fundamentals
Then open user data file
Copy the code & paste it here

<https://www.youtube.com/@techwithtwinkle>

☐ User data has already been base64 encoded

Summary

Number of Instances 1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.8.2...read more
ami-07f0dc6c4fe353ff1

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Cancel Launch instance Preview code

aws [Alt+S] Asia Pacific (Hyderabad) twinkie @ aws-twinkle-v4

EC2 > Instances > Launch an Instance

User data - optional [Info](#)
Upload a file with your user data or enter it in the field.

[Choose file](#)

```
#!/bin/bash
# Use this for your user data (script from top to bottom)
# install httpd (Linux 2 version)
yum update -y
yum install -y httpd
systemctl start httpd
systemctl enable httpd
echo "<h1>Hello World from ${hostname -f}</h1>" >
/var/www/html/index.html
```

☐ User data has already been base64 encoded

Summary

Number of instances [Info](#)
1

Software image (AMI)
Amazon Linux 2023 AMI 2023.8.2...[read more](#)
ami-07f0dc6c4fe353ff1

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

[Cancel](#) [Launch instance](#) [Preview code](#)

<https://www.youtube.com/@techwithtwinkle>

Click on Launch Instance

aws [Alt+S] Asia Pacific (Hyderabad) twinkie @ aws-twinkle-v4

EC2 > Instances

EC2

- Dashboard
- EC2 Global View
- Events
- ▼ Instances
 - Instances
 - Instance Types
 - Launch Templates
 - Spot Requests
 - Savings Plans
 - Reserved Instances
 - Dedicated Hosts
 - Capacity Reservations
- ▼ Images
 - AMIs

Instances (1/1) [Info](#) Last updated less than a minute ago [Refresh](#) [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Find Instance by attribute or tag (case-sensitive) All states < 1 >

☒	Name	Instance ID	Instance state	Instance type	Status check	Alarm status
☒	My First Server	i-0880238b68271661d	Running	t3.micro	Initializing	View alarms +

i-0880238b68271661d (My First Server)

[Details](#) [Status and alarms](#) [Monitoring](#) [Security](#) [Networking](#) [Storage](#) [Tags](#)

▼ **Instance summary** [Info](#)

Instance ID i-0880238b68271661d	Public IPv4 address 18.60.50.174 open address	Private IPv4 addresses 172.31.37.52
--	--	--

In instances you will see your created instance

EC2 > Instances

EC2

- Dashboard
- EC2 Global View
- Events
- Instances
 - Instances
 - Instance Types
 - Launch Templates
 - Spot Requests
 - Savings Plans
 - Reserved Instances
 - Dedicated Hosts
 - Capacity Reservations
- Images
 - AMIs

Find Instance by attribute or tag (case-sensitive)

All states

Name	Instance ID	Instance state	Instance type	Status check	Alarm status
i-0880238b68271661d (My First Server)					
Instance ID	i-0880238b68271661d	Public IPv4 address	18.60.50.174 open address	Private IPv4 addresses	172.31.37.52
IPv6 address		Instance state	Running	Public DNS	ec2-18-60-50-174.ap-south-2.compute.amazonaws.com

Copy this public IP & Paste it in browser

http://18.60.50.174

Type http:// before IP Address

Otherwise
you will get
this error



18.60.50.174 doesn't support a secure connection with HTTPS

You are seeing this warning because this site does not support HTTPS and you are in Incognito mode. [Learn more about this warning](#)

Continue to site

Go back

<https://www.youtube.com/@techwithtwinkle>

Hello World from ip-172-31-37-52.ap-south-2.compute.internal



Our Site is working now

<https://www.youtube.com/@techwithtwinkle>

Go to Instance

The screenshot shows the AWS Management Console interface. On the left, the navigation menu is visible with 'EC2' selected. The main content area displays the 'Instance summary for i-0880238b68271661d (My First Server)'. The instance is in the 'Running' state. An orange arrow points to the 'Instance state' button, which is labeled 'Click here'.

Instance ID	Public IPv4 address	Private IPv4 addresses
i-0880238b68271661d	18.60.50.174 open address	172.31.37.52

IPv6 address	Instance state	Public DNS
-	Running	ec2-18-60-50-174.ap-south-2.compute.amazonaws.com open address

Hostname type	Private IP DNS name (IPv4 only)	Elastic IP addresses
IP name: ip-172-31-37-52.ap-south-2.compute.internal	ip-172-31-37-52.ap-south-2.compute.internal	-

Answer private resource DNS name	Instance type
IPv4 (A)	t3.micro

aws [Search] [Alt+S] Asia Pacific (Hyderabad) twinkl@ aws-twinkl-v4

EC2 > Instances > i-0880238b68271661d

EC2

- Dashboard
- EC2 Global View
- Events
- ▼ Instances
 - Instances
 - Instance Types
 - Launch Templates
 - Spot Requests
 - Savings Plans
 - Reserved Instances
 - Dedicated Hosts
 - Capacity Reservations
- ▼ Images
 - AMIs

Instance summary for i-0880238b68271661d (My First Server) Info

Updated 8 minutes ago

[Refresh](#) [Connect](#) [Instance state ▲](#) [Actions ▼](#)

Instance ID: [i-0880238b68271661d](#)

IPv6 address: [\[icon\]](#)

Private IPv4 addresses: [\[icon\]](#) 172.31.37.52

Public DNS: [\[icon\]](#) ec2-18-60-50-174.ap-south-2.compute.amazonaws.com | [open address](#)

Private IP DNS name (IPv4 only): [\[icon\]](#) ip-172-31-37-52.ap-south-2.compute.internal

Instance type: t3.micro

Answer private resource DNS name: IPv4 (A)

Actions menu:

- Stop instance
- Start instance
- Reboot instance
- Hibernate instance
- Terminate (delete) instance

Click here to reboot Instance – If you want

Click here to stop Instance – If you want

Click here to delete Instance – If you want

EC2 Instance Types

You can check Instance type :- <https://aws.amazon.com/ec2/instance-types/>

Instance Family	Purpose	Description	Use Case Examples
General Purpose	Balanced CPU, memory, and networking	Ideal for diverse workloads	Web servers, dev/test, small databases
Compute Optimized	High-performance CPU	Designed for compute-intensive tasks	Gaming servers, batch processing, web servers
Memory Optimized	More memory per vCPU	Ideal for memory-intensive workloads	Large databases, in-memory caches
Storage Optimized	High-speed local storage	Designed for large, low-latency storage	Data warehousing, big data, log processing
Accelerated Computing	GPU or FPGA powered	Optimized for ML, HPC, video encoding	AI/ML training, 3D rendering, scientific computing

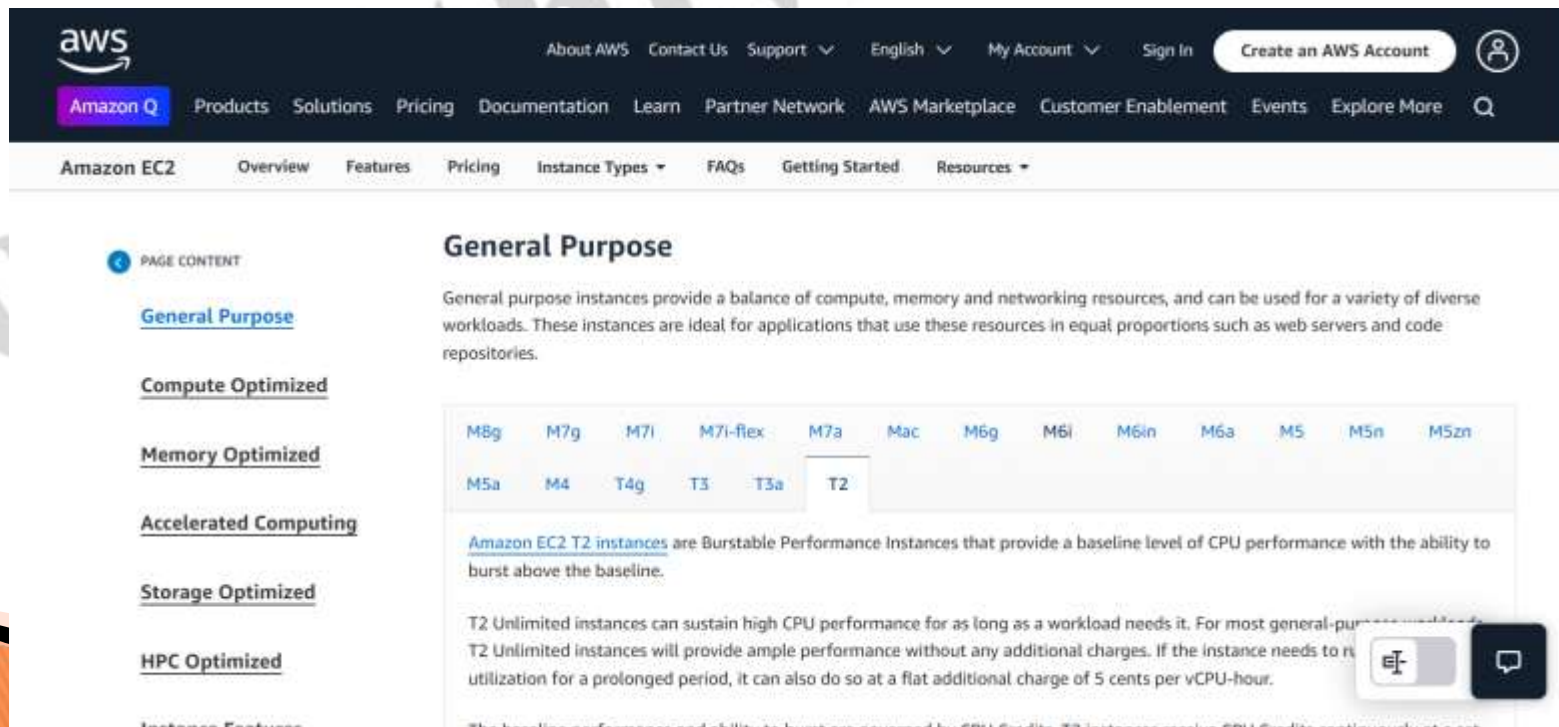
<https://www.youtube.com/@techwithtwinkle>

General Purpose

► Type

<https://www.youtube.com/@techwithtwinkle>

- t4g / t3 / t3a :- Low-cost burstable performance instances using credits. Best for intermittent workloads.
- m7g / m6g / m5 / m5a :- Balanced compute/memory. Suitable for apps needing predictable performance.



The screenshot shows the AWS General Purpose page. The top navigation bar includes the AWS logo, links for About AWS, Contact Us, Support, English, My Account, Sign In, and a button to Create an AWS Account. Below this is a secondary navigation bar with links for Amazon Q, Products, Solutions, Pricing, Documentation, Learn, Partner Network, AWS Marketplace, Customer Enablement, Events, and Explore More. The main content area is titled "General Purpose" and describes these instances as providing a balance of compute, memory, and networking resources. A table lists various instance types, with T2 highlighted. Below the table, text explains that Amazon EC2 T2 instances are burstable performance instances. The page also includes a sidebar with links to General Purpose, Compute Optimized, Memory Optimized, Accelerated Computing, Storage Optimized, and HPC Optimized.

General Purpose

General purpose instances provide a balance of compute, memory and networking resources, and can be used for a variety of diverse workloads. These instances are ideal for applications that use these resources in equal proportions such as web servers and code repositories.

M8g	M7g	M7i	M7i-flex	M7a	Mac	M6g	M6i	M6in	M6a	M5	M5n	M5zn
M5a	M4	T4g	T3	T3a	T2							

[Amazon EC2 T2 instances](#) are Burstable Performance Instances that provide a baseline level of CPU performance with the ability to burst above the baseline.

T2 Unlimited instances can sustain high CPU performance for as long as a workload needs it. For most general-purpose workloads, T2 Unlimited instances will provide ample performance without any additional charges. If the instance needs to sustain high utilization for a prolonged period, it can also do so at a flat additional charge of 5 cents per vCPU-hour.

Compute Optimized

Type

- ▶ c7g / c6g / c5 / c5n:- High ratio of compute power. Ideal for compute-bound apps (e.g., CPU-intensive workloads like video encoding or high-performance web servers).

Compute Optimized

Compute Optimized instances are ideal for compute bound applications that benefit from high performance processors. Instances belonging to this category are well suited for batch processing workloads, media transcoding, high performance web servers, high performance computing (HPC), scientific modeling, dedicated gaming servers and ad server engines, machine learning inference and other compute intensive applications.

C8g	C8gn	C7g	C7gn	C7i	C7i-flex	C7a	C6g	C6gn	C6i	C6in	C6a	C5
C5n	C5a	C4										

[Amazon EC2 C8g instances](#) are powered by AWS Graviton4 processors. They deliver the best price performance in Amazon EC2 for compute-intensive workloads.

<https://www.youtube.com/@techwithtwinkle>

Memory Optimized

Type

- ▶ r7g / r6g / r5 / r5n :- High memory-to-vCPU ratio. Good for in-memory DBs, real-time analytics.
- ▶ x2gd / x1e / x1 :- Extra-large memory instances for SAP HANA, high-performance DBs.
- ▶ Z1d :- High-frequency CPU and large memory for electronic design and gaming.

Memory Optimized

<https://www.youtube.com/@techwithtwinkle>

Memory optimized instances are designed to deliver fast performance for workloads that process large data sets in memory.

R8g	R7g	R7i	R7iz	R7a	R6g	R6i	R6in	R6a	R5	R5n	R5b	R5a	R4
U7i	High Memory (U-1)			X8g	X2gd	X2idn	X2iedn	X2iezn	X1	X1e	z1d		

[Amazon EC2 R8g instances](#) are powered by AWS Graviton4 processors. They deliver the best price performance in Amazon EC2 for memory-intensive workloads.

Accelerated Computing

Type

- ▶ p4 / p3 / g5 / g4ad :- GPU-powered. Designed for ML, gaming, video transcoding.
- ▶ inf2 / inf1 :- Inferentia-powered for ML inference at low cost.
- ▶ F1 :- FPGA-powered for custom hardware acceleration (e.g., genomics).

Accelerated Computing

<https://www.youtube.com/@techwithtwinkle>

Accelerated computing instances use hardware accelerators, or co-processors, to perform functions, such as floating point number calculations, graphics processing, or data pattern matching, more efficiently than is possible in software running on CPUs.

P6e	P6	P5	P4	G6e	G6	G5g	G5	G4dn	G4ad	Trn2	Trn1	Inf2	Inf1
DL1	DL2q	F2	VT1										

[Amazon EC2 P6e-GB200 UltraServers](#) accelerated by NVIDIA GB200 NVL72 offer the highest GPU AI training and inference performance in Amazon Elastic Compute Cloud (Amazon EC2).

Storage Optimized

Type

- ▶ i4i / i3 / i3en :- High IOPS SSD-backed storage. Ideal for NoSQL databases and transactional systems.
- ▶ d3 / d2 :- HDD-backed with massive storage. Great for data warehousing, Hadoop.

Storage Optimized <https://www.youtube.com/@techwithtwinkle>

Storage optimized instances are designed for workloads that require high, sequential read and write access to very large data sets on local storage. They are optimized to deliver millions of low-latency, random I/O operations per second (IOPS) to applications.

I8g	I7i	I7ie	I4g	Im4gn	Is4gen	I4i	I3	I3en	D3	D3en	D2	H1
------------	-----	------	-----	-------	--------	-----	----	------	----	------	----	----

[Amazon EC2 I8g instances](#) are powered by AWS Graviton4 processors and 3rd generation AWS Nitro SSDs. They deliver the best compute and storage performance among storage-optimized Amazon EC2 instances.

Tips for Choosing the Right Instance:

- ▶ **T-series (e.g., t3):** Best for low-cost, burstable workloads.
- ▶ **M-series (e.g., m5):** Go-to for general workloads if unsure.
- ▶ **C-series (e.g., c6g):** Choose when your app is CPU-heavy.
- ▶ **R/X-series:** Memory-bound workloads like Redis, SAP HANA.
- ▶ **I/D-series:** Use when you need fast local storage.
- ▶ **G/P-series:** When GPU or ML acceleration is required.

<https://www.youtube.com/@techwithtwinkle>

EC2 Instance Types: Example

Instance	vCPU	Mem (GiB)	Storage	Network Performance	EBS Bandwidth (Mbps)
t2.micro	1	1	EBS-Only	Low to Moderate	
t2.xlarge	4	16	EBS-Only	Moderate	
c5d.4xlarge	16	32	1 x 400 NVMe SSD	Up to 10 Gbps	4,750
r5.16xlarge	64	512	EBS Only	20 Gbps	13,600
m5.8xlarge	32	128	EBS Only	10 Gbps	6,800

t2.micro is part of the AWS free tier (up to 750 hours per month)

Great Website: <https://instances.vantage.sh/>

<https://www.youtube.com/@techwithtwinkle>

Security Groups

- ▶ A **Security Group (SG)** acts as a **virtual firewall** for your **EC2 instances** (or other resources like RDS, Lambda with VPC). It controls **inbound and outbound traffic at the instance level**.

Basic Properties of Security Groups

Feature	Description
Stateful	Responses to allowed inbound traffic are automatically allowed outbound.
Instance-level	Attached directly to EC2 instances, not subnets (unlike NACLs).
Only Allow Rules	You can only allow traffic, not explicitly deny it.
Rules Based on IP / SG / Ports	Rules are defined using protocols (e.g., TCP), port ranges, and source/destination IPs or other SGs.

<https://www.youtube.com/@techwithtwinkle>

Types of Rules

▶ Inbound Rules

- Define what traffic **can come in** to the instance.
- Example
- Allow TCP port 22 from 203.0.113.10/32 :- SSH access only from a specific IP
- Allow TCP port 80 from 0.0.0.0/0 :- Public HTTP access from anywhere

▶ Outbound Rules

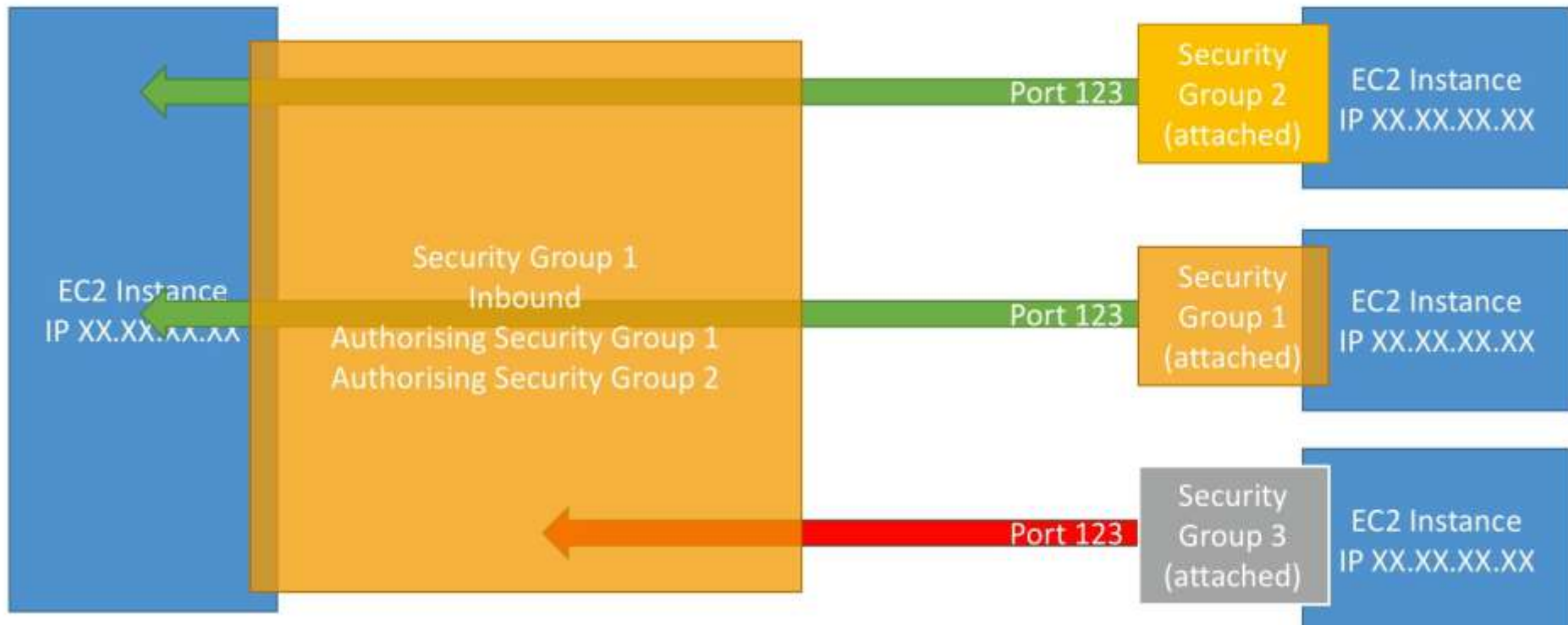
- Define what traffic **can leave** the instance.
- Example
- Allow all traffic (default) :- Most SGs allow all outbound traffic by default
- Allow TCP port 443 to 0.0.0.0/0 :- Outbound HTTPS access

Direction	Type	Protocol	Port	Source/Destination
Inbound	HTTP	TCP	80	0.0.0.0/0
Inbound	SSH	TCP	22	Your IP
Outbound	ALL	ALL	ALL	0.0.0.0/0

Security Groups Diagram



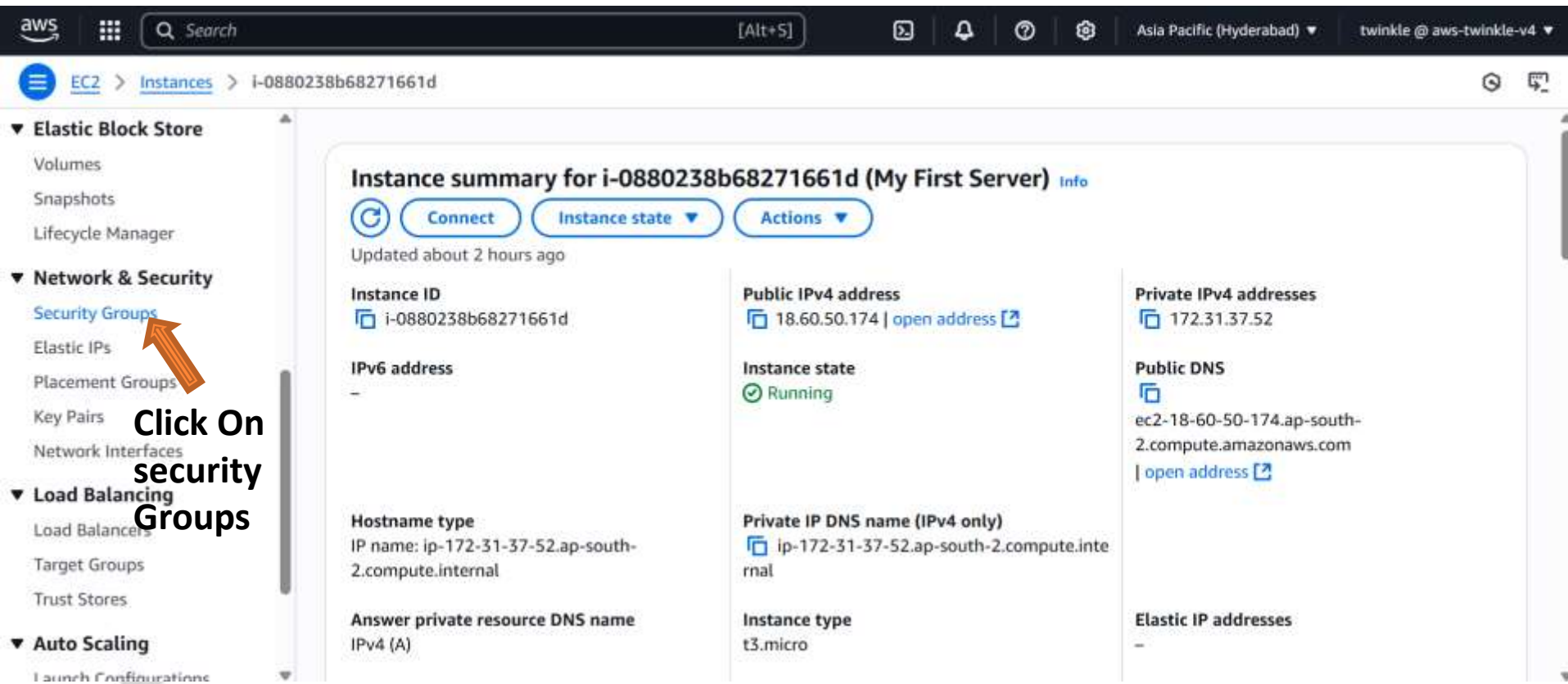
Referencing Other security Groups



Classic Ports to Know

- ▶ 22 – SSH (Secure Shell) – Log into a Linux instance
- ▶ 21 = FTP (File Transfer Protocol) – upload files into a file share
- ▶ 22 = SFTP (Secure File Transfer Protocol) – upload files using SSH
- ▶ 80 = HTTP – access unsecured websites
- ▶ 443 = HTTPS – access secured websites
- ▶ 3389 = RDP (Remote Desktop Protocol) – log into a Windows Instance

Security Groups Hands On



The screenshot displays the AWS Management Console interface. The top navigation bar includes the AWS logo, a search bar, and the user's profile. The left-hand navigation pane is expanded, showing the 'Network & Security' section. An orange arrow points to the 'Security Groups' link. The main content area shows the 'Instance summary for i-0880238b68271661d (My First Server)'. The instance is in a 'Running' state. The summary includes details such as the Instance ID, Public IPv4 address (18.60.50.174), Private IPv4 addresses (172.31.37.52), Public DNS (ec2-18-60-50-174.ap-south-2.compute.amazonaws.com), Private IP DNS name (ip-172-31-37-52.ap-south-2.compute.internal), and Instance type (t3.micro).

Instance summary for i-0880238b68271661d (My First Server) Info

Updated about 2 hours ago

Instance ID
i-0880238b68271661d

Public IPv4 address
18.60.50.174 | [open address](#)

Private IPv4 addresses
172.31.37.52

Instance state
Running

Public DNS
ec2-18-60-50-174.ap-south-2.compute.amazonaws.com | [open address](#)

IPv6 address
-

Private IP DNS name (IPv4 only)
ip-172-31-37-52.ap-south-2.compute.internal

Instance type
t3.micro

Hostname type
IP name: ip-172-31-37-52.ap-south-2.compute.internal

Answer private resource DNS name
IPv4 (A)

Elastic IP addresses
-

Security Groups

Network & Security

Load Balancing

Auto Scaling

<https://www.youtube.com/@techwithtwinkle>

aws [Search] [Alt+S] Asia Pacific (Hyderabad) twinkie @ aws-twinkle-v4

EC2 > Security Groups

Security Groups (2) info

Find security groups by attribute or tag

☐	Name	Security group ID	Security group name	VPC ID
☐	-	sg-03fd323ae6499176c	default	vpc-04c59650b26f4b57f
☐	-	sg-080ecfdb5e3186c8	launch-wizard-1	vpc-04c59650b26f4b57f

Select security group

This is your default security group

**This one is our EC2 instance created security group
Click on this group**

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EC2 > Security Groups > sg-080ecfdb5e3186c8 - launch-wizard-1

Elastic Block Store
Volumes
Snapshots
Lifecycle Manager

Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces

Load Balancing
Load Balancers
Target Groups
Trust Stores

Auto Scaling
Launch Configurations

Security group name
[launch-wizard-1](#)

Security group ID
[sg-080ecfdb5e3186c8](#)

Description
[launch-wizard-1](#) created 2025-07-19T06:04:43.525Z

VPC ID
[vpc-04c59650b26f4b57f](#)

Owner
[235562991793](#)

Inbound rules count
2 Permission entries

Outbound rules count
1 Permission entry

<https://www.youtube.com/@techwithtwinkle>

Here is your inbound rule

**On clicking edit inbound rule
You can edit rule**

Inbound rules (2)

Search

☐	Name	Security group rule ID	IP version	Type	Protocol
☐	-	sgr-01331c9ce0e81c0e7	IPv4	SSH	TCP
☐	-	sgr-0c4e8d7b9adcfa794	IPv4	HTTP	TCP

aws [Search] [Alt+S] Asia Pacific (Hyderabad) twinkie @ aws-twinkle-v4

EC2 > Security Groups > sg-080ecfdb5e3186c8 - launch-wizard-1 > Edit inbound rules

Edit inbound rules Info

Inbound rules control the incoming traffic that's allowed to reach the instance.

Security group rule ID	Type Info	Protocol Info	Port range Info	Source Info	Description - optional Info	
sgr-01331c9ce0e81c0e7	SSH	TCP	22	Cus...		Delete
sgr-0c4e8d7b9adcfa794	HTTP	TCP	80	Cus...		Delete

[Add rule](#) You can add rule here, if you wanna add

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses on

Cancel [Preview changes](#) [Save rules](#)

After that click on save rules

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EC2 > Security Groups > sg-080ecfdb5e3186c8 - launch-wizard-1

sg-080ecfdb5e3186c8 - launch-wizard-1

Details

Security group name launch-wizard-1	Security group ID sg-080ecfdb5e3186c8	Description launch-wizard-1 created 2025-07-19T06:04:43.525Z	VPC ID vpc-04c59650b26f4b576
Owner 235562991793	Inbound rules count 2 Permission entries	Outbound rules count 1 Permission entry	

[Inbound rules](#) [Outbound rules](#) [Sharing - new](#) [VPC associations - new](#) [Tags](#)

Outbound rules (1)

Name	Security group rule ID	IP version	Type	Protocol
-	sgr-00cbd7f3b0ddffba7	IPv4	All traffic	All

[Click on outbound rules](#)

[Click on edit outbound rules](#)

aws Search [Alt+S] Asia Pacific (Hyderabad) twinkie @ aws-twinkle-v4

EC2 > Security Groups > sg-080ecfdb5e3186c8 - launch-wizard-1 > Edit outbound rules

Edit outbound rules

Outbound rules control the outgoing traffic that's allowed to leave the instance.

Outbound rules

Security group rule ID	Type	Protocol	Port range	Destination	Description - optional	
sg-00cbd7f3b0ddffba7	All traffic	All	All	Cus... 0.0.0.0/0		Delete

Add rule Click on add rule as you want to add

At last click on save rules

Rules with destination of 0.0.0.0/0 or ::/0 allow your instances to send traffic to any IPv4 or IPv6 address. We recommend setting security group rules to be more restrictive and to only allow traffic to specific known IP addresses.

Cancel Preview changes Save rules

Here you will get to know that how you can edit security groups Rule

<https://www.youtube.com/@techwithtwinkle>

SSH

- ▶ **SSH (Secure Shell)** is a **network protocol** that allows you to securely connect to another computer (typically a Linux server) over an unsecured network.
- ▶ It uses **encryption** to secure the connection.
- ▶ Most commonly used to **remotely log in to Linux servers**.
- ▶ Default port: **22**

<https://www.youtube.com/@techwithtwinkle>

	SSH	Putty	EC2 Instance Connect
Mac	✓		✓
Linux	✓		✓
Windows < 10		✓	✓
Windows >= 10	✓	✓	✓

How to SSH using Linux or Mac

The screenshot shows the AWS Management Console interface for EC2 instances. The left sidebar contains navigation links for EC2, Dashboard, EC2 Global View, Events, Instances, Instance Types, Launch Templates, Spot Requests, Savings Plans, Reserved Instances, Dedicated Hosts, Capacity Reservations, Images, and AMIs. The main content area displays a list of instances with columns for Name, Instance ID, Instance state, Instance type, Status check, and Alarm status. The instance 'My First Server' (ID: i-0880238b68271661d) is shown in the 'Running' state. Below the list, the details for this instance are shown, including the public IPv4 address 18.60.50.174, which is highlighted with an arrow and the text 'Copy your public IP Address'.

Instances (1/1) Info Last updated less than a minute ago [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Find Instance by attribute or tag (case-sensitive) All states

Name	Instance ID	Instance state	Instance type	Status check	Alarm status
My First Server	i-0880238b68271661d	Running	t3.micro	3/3 checks passed	View alarms

i-0880238b68271661d (My First Server)

[Details](#) [Status and alarms](#) [Monitoring](#) [Security](#) [Networking](#) [Storage](#) [Tags](#)

Instance summary Info

Instance ID	Public IPv4 address	Private IPv4 addresses
i-0880238b68271661d	18.60.50.174 open address	172.31.37.52

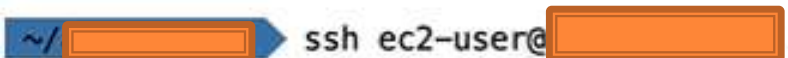
<https://www.youtube.com/@techwithtwinkle>



Type this &
paste your
public IP
address



Go to your linux or Mac console



```
~ / ssh ec2-user@  
Warning: Permanently added (ED25519) to the list of known hosts.  
Received disconnect from port 22:2: Too many authentication failures  
Disconnected from port 22
```



After that this Error will show here

<https://www.youtube.com/@techwithtwinkle>

- ▶ Download a keypair file which is with extension .pem of your instance

```
ls  
EC2Tutorial.pem archive
```

If you type ls it will show your file Make sure your file has no space in its name before this make sure you are in correct directory

```
~ cd aws-course  
pwd  
/Users/  
~/aws-course ls  
EC2Tutorial.pem archive
```

Here paste your public IP

```
ssh -i EC2Tutorial.pem ec2-user@  
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@  
@      WARNING: UNPROTECTED PRIVATE KEY FILE!      @  
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@  
Permissions 0644 for 'EC2Tutorial.pem' are too open.  
It is required that your private key files are NOT accessible by others.  
This private key will be ignored.  
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@  
@      WARNING: UNPROTECTED PRIVATE KEY FILE!      @  
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@  
Permissions 0644 for 'EC2Tutorial.pem' are too open.  
It is required that your private key files are NOT accessible by others.  
This private key will be ignored.  
Load key "EC2Tutorial.pem": bad permissions  
Received disconnect from 3.250.26.200 port 22:2: Too many authentication failures  
Disconnected from 3.250.26.200 port 22
```

You will see this error

```
@      WARNING: UNPROTECTED PRIVATE KEY FILE!      @
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
Permissions 0644 for 'EC2Tutorial.pem' are too open.
It is required that your private key files are NOT accessible by others.
This private key will be ignored.
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
@      WARNING: UNPROTECTED PRIVATE KEY FILE!      @
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
Permissions 0644 for 'EC2Tutorial.pem' are too open.
It is required that your private key files are NOT accessible by others.
This private key will be ignored.
Load key "EC2Tutorial.pem": bad permissions
Received disconnect from 3.250.26.200 port 22:2: Too many authentication failures
Disconnected from 3.250.26.200 port 22
~ chmod 0400 EC2Tutorial.pem
```

Enter this to remove above error
After that type clear & press enter

```
~ ssh -i EC2Tutorial.pem ec2-user@
Last login: Mon May 16 16:55:35 2022 from ec2-18-202-216-50.eu-west-1.compute.amazonaws.com

 _ | _ | _ )
 _ | ( _ /   Amazon Linux 2 AMI
 _ | \ _ | _ |

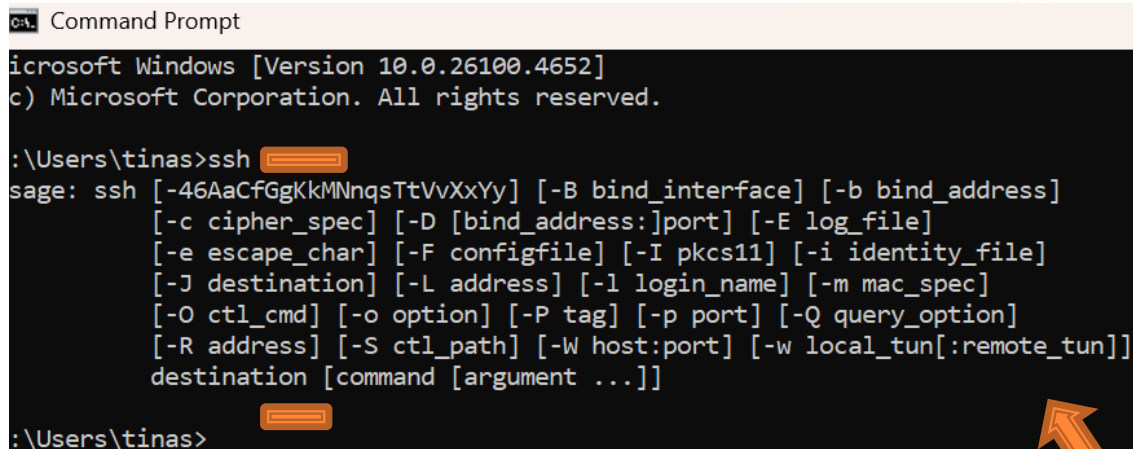
https://aws.amazon.com/amazon-linux-2/
-bash: warning: setlocale: LC_CTYPE: cannot change locale (UTF-8): No such file or directory
[ec2-user@ip-172-31-33-135 ~]$
```

Type same thing with public IP you will login to
your account

<https://www.youtube.com/@techwithtwinkle>

How to SSh Using Windows 10 or above

- ▶ Go to command prompt or windows PowerShell type SSH



Command Prompt

Microsoft Windows [Version 10.0.26100.4652]
c) Microsoft Corporation. All rights reserved.

C:\Users\tinas>ssh

Usage: ssh [-46AaCfGgKkMMNnqsTtVvXxYy] [-B bind_interface] [-b bind_address]
[-c cipher_spec] [-D [bind_address:]port] [-E log_file]
[-e escape_char] [-F configfile] [-I pkcs11] [-i identity_file]
[-J destination] [-L address] [-l login_name] [-m mac_spec]
[-O ctl_cmd] [-o option] [-P tag] [-p port] [-Q query_option]
[-R address] [-S ctl_path] [-W host:port] [-w local_tun[:remote_tun]]
destination [command [argument ...]]

C:\Users\tinas>

This will show

<https://www.youtube.com/@techwithtwinkle>

Windows PowerShell

Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! <https://aka.ms/PSWindows>

PS C:\Users\[redacted]> ssh

```
usage: ssh [-46AaCfGgKkMnNqsTtVvXxYy] [-B bind_interface] [-b bind_address]
[-c cipher_spec] [-D [bind_address:]port] [-E log_file]
[-e escape_char] [-F configfile] [-I pkcs11] [-i identity_file]
[-J destination] [-L address] [-l login_name] [-m mac_spec]
[-O ctl_cmd] [-o option] [-P tag] [-p port] [-Q query_option]
[-R address] [-S ctl_path] [-W host:port] [-w local_tun[:remote_tun]]
destination [command [argument ...]]
```

PS C:\Users\[redacted]>

**Same in PowerShell**

<https://www.youtube.com/@techwithtwinkle>

```
Windows PowerShell
PS C:\Users\tinas> ls

Directory: C:\Users\tinas

Mode                LastWriteTime         Length Name
----                -
d-----          30-03-2020         18:53      .android
d-----          30-08-2022         15:24      .aws
d-----          30-03-2020         18:52      .BigNox
d-----          08-07-2022         13:12      .openjfx
d-----          08-07-2022         13:12      .ScreamingFrogSEOSpider
d-r-----        29-03-2021         20:55      3D Objects
d-r-----        15-08-2024         12:08      Cisco Packet Tracer 8.2.1
d-r-----        20-01-2025         14:09      Contacts
d-r-----        30-10-2019         13:52      Documents
d-r-----        19-07-2025         14:55      Downloads
d-r-----        20-01-2025         14:09      Favorites
d-r-----        29-07-2024         13:50      GNS3
d-r-----        20-01-2025         14:09      Links
d-r-----        20-01-2025         14:09      Music
d-r-----        02-03-2020         11:43      Nox_share
d-r-----        20-01-2025         14:04      OneDrive
d-r-----        05-09-2019         08:04      Roaming
d-r-----        20-01-2025         14:09      Saved Games
d-r-----        20-01-2025         14:09      Searches
d-r-----        15-04-2025         15:50      Telegram Desktop
d-r-----        20-01-2025         14:09      Videos
d-r-----        30-03-2020         18:53      .vscode
```

First we will go
to our directory
where the file
exist,
For now my file
is exist at
downloads

```
PS C:\Users\tinas> cd .\Downloads
PS C:\Users\tinas\Downloads>
```

Directory Changed



Type this with your public IP & keypair file name



Type yes & Enter

<https://www.youtube.com/@techwithtwinkle>

SSH Troubleshooting

1) There's a connection timeout

This is a security group issue. Any timeout (not just for SSH) is related to security groups or a firewall. Ensure your security group looks like this and correctly assigned to your EC2 instance.



The screenshot shows the AWS Management Console interface for a Security Group. The 'Inbound' tab is selected, displaying a table of inbound rules. The table has four columns: Type, Protocol, Port Range, and Source. There are two rules listed, both for SSH using TCP on port 22. The first rule has a source of 0.0.0.0/0, and the second rule has a source of ::/0.

Type	Protocol	Port Range	Source
SSH	TCP	22	0.0.0.0/0
SSH	TCP	22	::/0

2) There's still a connection timeout issue

If your security group is properly configured as above, and you still have connection timeout issues, then that means a corporate firewall or a personal firewall is blocking the connection. **Please use EC2 Instance Connect as described in the next lecture**

3) SSH does not work on Windows

If it says: ssh command not found, that means you have to use Putty

Follow again the video. If things don't work, please use EC2 Instance Connect as described in the next lecture.

<https://www.youtube.com/@techwithtwinkle>

4) There's a connection refused

This means the instance is reachable, but no SSH utility is running on the instance

Try to restart the instance

If it doesn't work, terminate the instance and create a new one. Make sure you're using **Amazon Linux 2**

5) Permission denied (publickey,gssapi-keyex,gssapi-with-mic)

This means either two things:

You are using the wrong security key or not using a security key. Please look at your EC2 instance configuration to make sure you have assigned the correct key to it.

You are using the wrong user. Make sure you have started an **Amazon Linux 2 EC2 instance**, and make sure you're using the user **ec2-user**. This is something you specify when doing **ec2-user@<public-ip>** (ex: **ec2-user@18.60.50.174**) in your SSH command or your Putty configuration

6) Nothing is working - "aaaahhhhhh"

Don't panic. Use **EC2 Instance Connect** from the next lecture. Make sure you started an **Amazon Linux 2** and you will be able to follow along with the tutorial :)

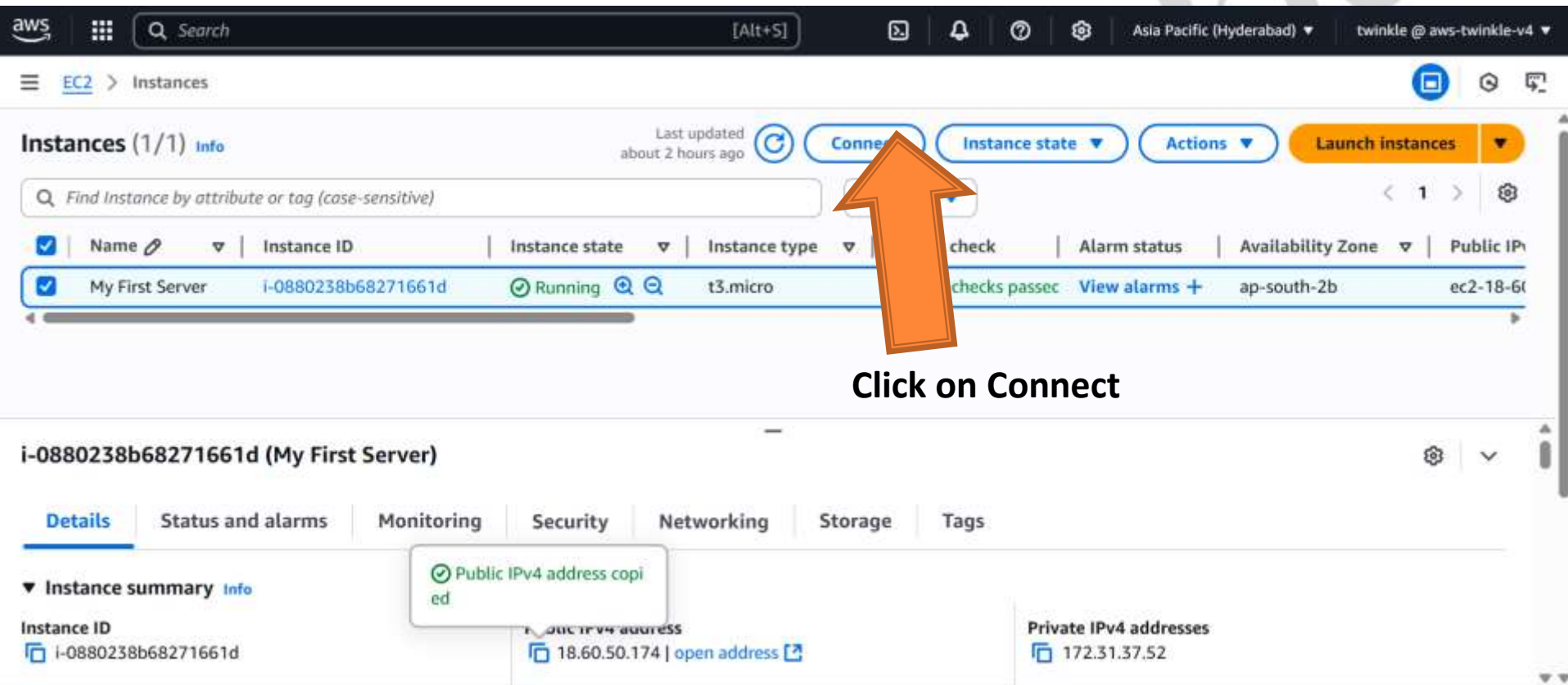
7) I was able to connect yesterday, but today I can't

This is probably because you have stopped your EC2 instance and then started it again today. **When you do so, the public IP of your EC2 instance will change.** Therefore, in your command, or Putty configuration, please make sure to edit and save the new public IP.

<https://www.youtube.com/@techwithtwinkle>

EC2 Connect

► Go to Instance



The screenshot shows the AWS Management Console interface for the EC2 Instances page. At the top, there's a search bar and navigation tabs. Below the navigation, the 'Instances (1/1)' section is visible. A table lists the instances, with the first instance 'My First Server' (ID: i-0880238b68271661d) in a 'Running' state. Above the table, there are buttons for 'Connect', 'Instance state', and 'Actions'. An orange arrow points to the 'Connect' button. Below the table, the details for the selected instance are shown, including the 'Instance summary' tab with the instance ID and public/private IP addresses.

aws [Search] [Alt+S] Asia Pacific (Hyderabad) twinkie @ aws-twinkle-v4

EC2 > Instances

Instances (1/1) Info Last updated about 2 hours ago

Find Instance by attribute or tag (case-sensitive)

Name	Instance ID	Instance state	Instance type	check	Alarm status	Availability Zone	Public IP
My First Server	i-0880238b68271661d	Running	t3.micro	checks passed	View alarms +	ap-south-2b	ec2-18-60-50-174

Click on Connect

i-0880238b68271661d (My First Server)

Details Status and alarms Monitoring Security Networking Storage Tags

Instance summary Info

Instance ID i-0880238b68271661d

Public IPv4 address 18.60.50.174 open address

Private IPv4 addresses 172.31.37.52

aws [Alt+S] Search Asia Pacific (Hyderabad) twinkl @ aws-twinkl-v4

EC2 > Instances > i-0880238b68271661d > Connect to instance

EC2 Instance Connect Session Manager SSH client EC2 serial console

Instance ID
i-0880238b68271661d (My First Server)

☒ Connect using a Public IP
Connect using a public IPv4 or IPv6 address

☐ Connect using a Private IP
Connect using a private IP address and a VPC endpoint

☒ Public IPv4 address
18.60.50.174

☐ IPv6 address

Username
Enter the username defined in the AMI used to launch the instance. If you didn't define a custom username, use the default username, ec2-user.

ec2-user

Note: In most cases, the default username, ec2-user, is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

Click on Connect

<https://www.youtube.com/@techwithtwinkle>

Cancel Connect

aws [Alt+S] Search Asia Pacific (Hyderabad) twinkl @ aws-twinkl-v4

```
A newer release of "Amazon Linux" is available.
Version 2023.0.20250715:
Run "/usr/bin/dnf check-release-update" for full release and version update info

Amazon Linux 2023

https://aws.amazon.com/linux/amazon-linux-2023

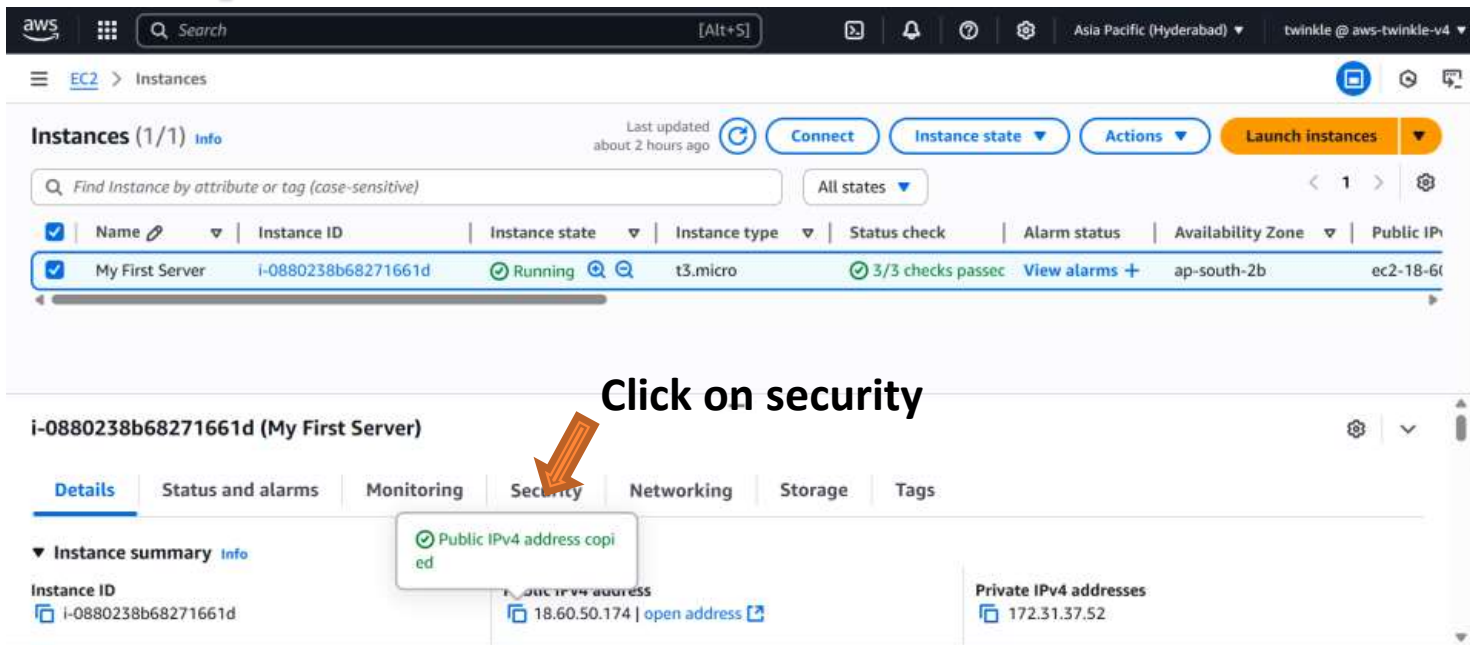
Last login: Sat Jul 19 11:12:57 2025 from 103.70.167.59
[ec2-user@ip-172-31-37-52 ~]$ whoami
ec2-user
[ec2-user@ip-172-31-37-52 ~]$ ping google.com
PING google.com (142.251.43.46) 56(84) bytes of data:
64 bytes from bkk02s01-in-f14.1e100.net (142.251.43.46): icmp_seq=1 ttl=116 time=31.4 ms
64 bytes from bkk02s01-in-f14.1e100.net (142.251.43.46): icmp_seq=2 ttl=116 time=31.4 ms
64 bytes from bkk02s01-in-f14.1e100.net (142.251.43.46): icmp_seq=3 ttl=116 time=31.4 ms
64 bytes from bkk02s01-in-f14.1e100.net (142.251.43.46): icmp_seq=4 ttl=116 time=31.4 ms
64 bytes from bkk02s01-in-f14.1e100.net (142.251.43.46): icmp_seq=5 ttl=116 time=31.4 ms
```

Our EC2 server working
Check it by ping google.com you can see in this image

i-0880238b68271661d (My First Server)

PublicIPs: 18.60.50.174 PrivateIPs: 172.31.37.52

If you have IPv6 Address



The screenshot shows the AWS Management Console 'Instances' page. The instance 'My First Server' (ID: i-0880238b68271661d) is in a 'Running' state. The 'Security' tab is selected, and an orange arrow points to it with the text 'Click on security'. Below the tabs, the 'Instance summary' section shows the 'Instance ID' as i-0880238b68271661d. A tooltip indicates 'Public IPv4 address copied'.

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4
My First Server	i-0880238b68271661d	Running	t3.micro	3/3 checks passed	View alarms +	ap-south-2b	ec2-18-60-

i-0880238b68271661d (My First Server)

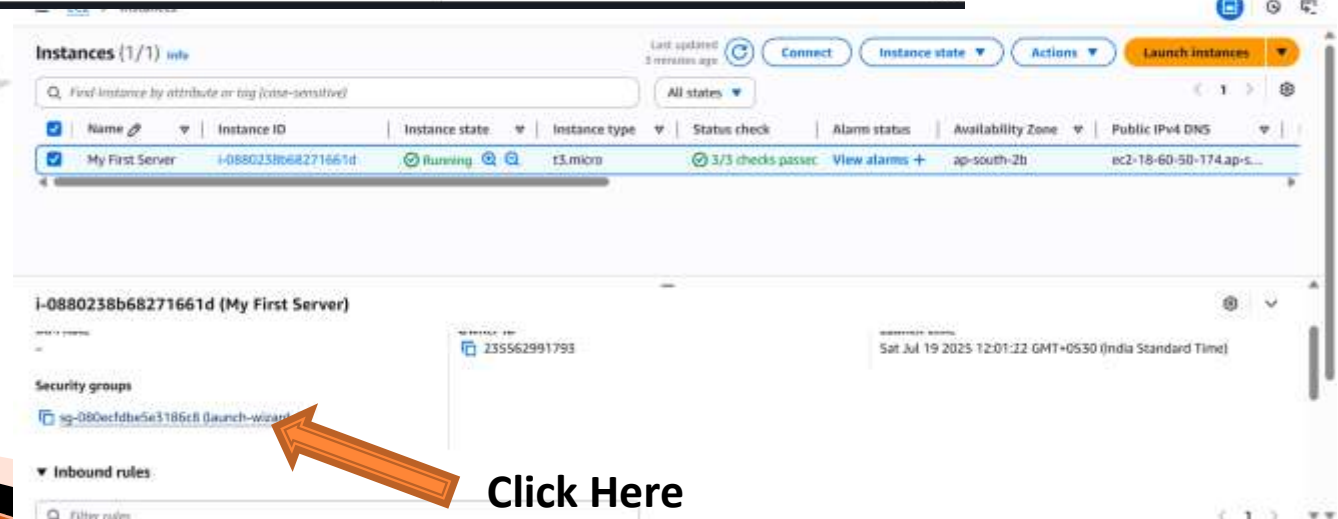
Details | Status and alarms | Monitoring | **Security** | Networking | Storage | Tags

▼ Instance summary Info

Instance ID
i-0880238b68271661d

Public IPv4 address
18.60.50.174 | open address

Private IPv4 addresses
172.31.37.52



The screenshot shows the 'Instances' page with the instance 'My First Server' (ID: i-0880238b68271661d) in a 'Running' state. The 'Security groups' section is expanded, and an orange arrow points to the 'Inbound rules' section with the text 'Click Here'.

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS
My First Server	i-0880238b68271661d	Running	t3.micro	3/3 checks passed	View alarms +	ap-south-2b	ec2-18-60-50-174.ap-s...

i-0880238b68271661d (My First Server)

Security groups
sg-080cfd8a5e3186e8 (launch-wizard-1)

▼ Inbound rules

Filter rules

sg-080ecfdb5e3186c8 - launch-wizard-1

Details

Security group name launch-wizard-1	Security group ID sg-080ecfdb5e3186c8	Description launch-wizard-1 created 2025-07-19T06:04:43.525Z.	VPC ID vpc-04c59650b26f4b576
Owner 235562991793	Inbound rules count 2 Permission entries	Outbound rules count 1 Permission entry	

Inbound rules (2)

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source
-	sgr-01331c9ce0e81c0e7	IPv4	SSH	TCP	22	0.0.0.0/0
-	sgr-0c4e8d7b9adcfa794	IPv4	HTTP	TCP	80	0.0.0.0/0

Click on Edit Inbound Rule

Edit inbound rules

Inbound rules control the incoming traffic that's allowed to reach the instance.

<https://www.youtube.com/@techwithtwinkle>

Inbound rules

Security group rule ID	Type	Protocol	Port range	Source	Description - optional
sgr-01331c9ce0e81c0e7	SSH	TCP	22	Cust... 0.0.0.0/0	Delete
sgr-0c4e8d7b9adcfa794	HTTP	TCP	80	Cust... 0.0.0.0/0	Delete

Add rule

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Cancel Preview changes Save rules

Click on Add Rule

aws [Search] [Alt+S] Asia Pacific (Hyderabad) twinkie @ aws-twinkle-v4

EC2 > Security Groups > sg-080ecfdb5e3186c8 - launch-wizard-1 > Edit inbound rules

Edit inbound rules Info

Inbound rules control the incoming traffic that's allowed to reach the instance.

Inbound rules Info

Security group rule ID	Type Info	Protocol Info	Port range Info	Source Info	Description - optional Info	
sgr-01331c9ce0e81c0e7	SSH	TCP	22	Custom...	0.0.0.0/0	Delete
sgr-0c4e8d7b9adcfa794	HTTP	TCP	80	Custom...	0.0.0.0/0	Delete
-	SSH	TCP	22	Custom...	0.0.0.0/0	Delete

[Add rule](#)

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting up rules to allow access from known IP addresses only.

[Preview changes](#) [Save rules](#)

Select SSH

Select Anywhere IPv6

Click on Save Rules

aws [Search] [Alt+S] Asia Pacific (Hyderabad) twinkie @ aws-twinkle-v4

EC2 > Security Groups > sg-080ecfdb5e3186c8 - launch-wizard-1

Inbound security group rules successfully modified on security group (sg-080ecfdb5e3186c8 | launch-wizard-1)

Details

Security group name	launch-wizard-1	Security group ID	sg-080ecfdb5e3186c8	Description	launch-wizard-1 created 2025-07-19T06:04:43.525Z	VPC ID	vpc-04c59650b26f4b576
Owner	235562991793	Inbound rules count	3 Permission entries	Outbound rules count	1 Permission entry		

[Inbound rules](#) [Outbound rules](#) [Sharing - new](#) [VPC associations - new](#) [Tags](#)

<https://www.youtube.com/@techwithtwinkle>

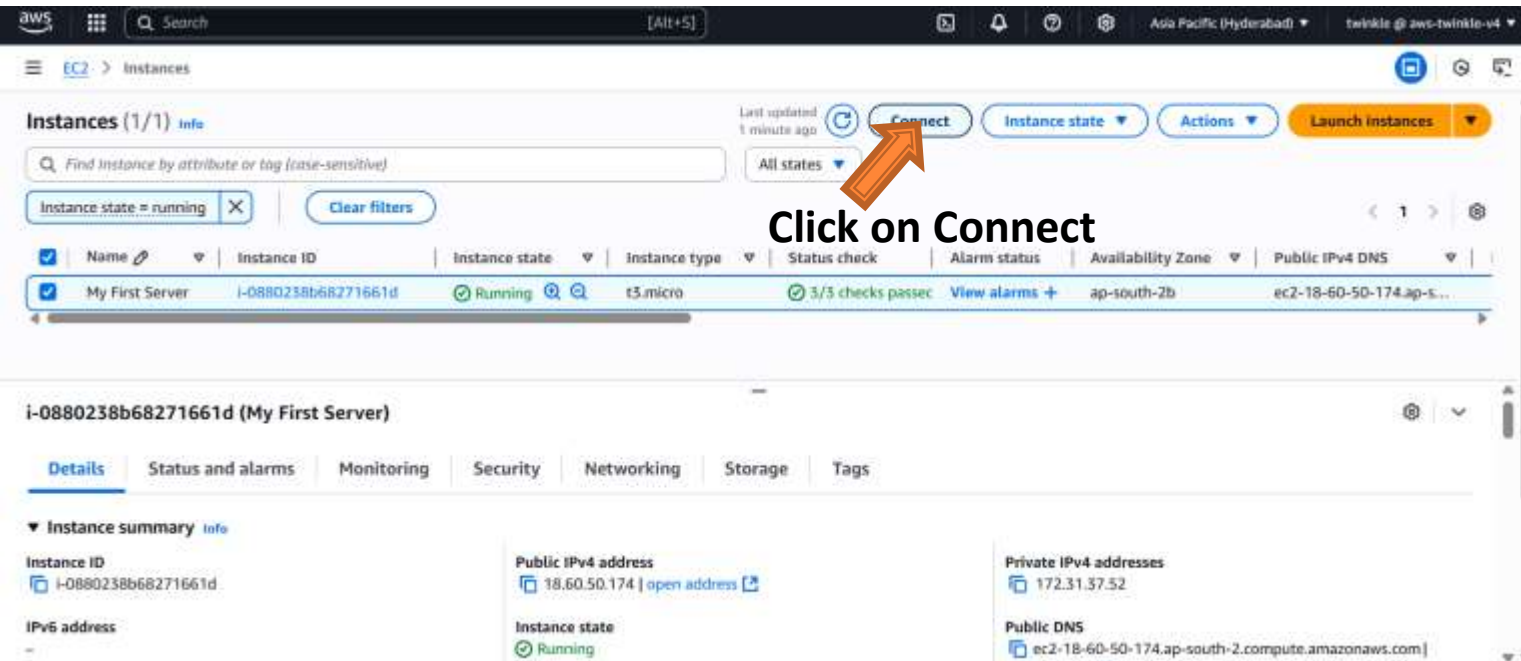
Inbound rules (3)

[Manage tags](#) [Edit inbound rules](#)

	Name	Security group rule ID	IP version	Type	Protocol	Port range	Source
☐	-	sgr-01331c9ce0e81c0e7	IPv4	SSH	TCP	22	0.0.0.0/0
☐	-	sgr-0e97b225e4969b8cf	IPv6	SSH	TCP	22	::/0
☐	-	sgr-0c4e8d7b9adcfa794	IPv4	HTTP	TCP	80	0.0.0.0/0

IPv6 Added

Go back to instance



The screenshot shows the AWS Management Console interface for EC2 instances. At the top, there's a search bar and navigation tabs. The 'Instances' tab is selected, showing a list of instances. The instance 'My First Server' (ID: i-0880238b68271661d) is in the 'Running' state. An orange arrow points to the 'Connect' button in the top right of the instance list. Below the list, the details for 'My First Server' are shown, including its public IPv4 address (18.60.50.174), private IPv4 address (172.31.37.52), and public DNS (ec2-18-60-50-174.ap-south-2.compute.amazonaws.com).

Click on Connect

Same you can do as IPv4 your EC2 Connect will work

<https://www.youtube.com/@techwithtwinkle>

EC2 Instance Roles Demo

```
aws
[Alt+S]
Asia Pacific (Hyderabad)
twinkle @ aws-twinkle-v4

A newer release of "Amazon Linux" is available.
Version 2023.8.20250715:
Run "/usr/bin/dnf check-release-update" for full release and version update info

Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

Last login: Sat Jul 19 11:26:51 2025 from 18.60.252.248
[ec2-user@ip-172-31-37-52 ~]$ aws --version
aws-cli/2.25.0 Python/3.9.23 Linux/6.1.141-165.249.amzn2023.x86_64 source/x86_64.amzn.2023
[ec2-user@ip-172-31-37-52 ~]$ aws iam list-users
Unable to locate credentials. You can configure credentials by running "aws configure"
[ec2-user@ip-172-31-37-52 ~]$ aws configure
AWS Access Key ID [None]:
```

As for AWS Configure you need to provide secret key which is not good practice so now we connect iam user to ec2



i-0880238b68271661d (My First Server)
PublicIPs: 18.60.50.174 PrivateIPs: 172.31.37.52

As previously we are already created IAM user with Readonly access I am connecting that user to EC2

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Identity and Access Management (IAM)

Roles (3) Info

An IAM role is an identity you can create that has specific permissions with credentials that are valid for short durations. Roles can be assumed by entities that you trust.

Search

☐	Role name	Trusted entities	Last activity
☐	AWSServiceRoleForSupport	AWS Service: support (Service-Linker	
☐	AWSServiceRoleForTrustedAdvisor	AWS Service: trustedadvisor (Service	
☐	DemoroleEC2	AWS Service: ec2	

Roles Anywhere Info

Authenticate your non AWS workloads and securely provide access to AWS services.

Manage

Go to EC2 instance

Here is our user with IAMreadonlyaccess permissions role

EC2 > Instances

Find Instance by attribute or tag (case-sensitive)

Instance state = running

Clear filters

☒	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS
	i-0880238b68271661d (My First Server)							
	IAM Role	-						Auto Scaling Group name
	IMDSv2	Required						Managed
	Operator	-						false

Instance ARN

arn:aws:ec2:ap-south-2:235562991793:instance/i-0880238b68271661d

Here is no iam user right now

<https://www.youtube.com/@techwithtwinkle>

Instances (1/1) Info

Find Instance by attribute or tag (case-sensitive)

Instance state = running

Clear filters

Last updated
9 minutes ago

Connect

Instance state

Actions

Launch instances

All states

Click on Actions

☒	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS
☒	My First Server	i-0880238b68271661d	Running	t3.micro	3/3 checks passed	View alarms +	ap-south-2b	ec2-18-60-50-174.ap-s...

i-0880238b68271661d (My First Server)

IAM Role

IMDSv2
Required

Operator

Subnet ID

subnet-00ce35864b297af51

Instance ARN

arn:aws:ec2:ap-south-2:235562991793:instance/i-0880238b68271661d

Auto Scaling Group name

Managed
false

<https://www.youtube.com/@techwithtwinkle>

Instances (1/1) Info

Find Instance by attribute or tag (case-sensitive)

Instance state = running

Clear filters

Last updated
10 minutes ago

Connect

Instance state

Actions

Launch instances

All states

- Instance diagnostics
- Instance settings
- Networking
- Security
- Image and templates
- Monitor and troubleshoot

Click on Security

☒	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone
☒	My First Server	i-0880238b68271661d	Running	t3.micro	3/3 checks passed	View alarms +	ap-south-2b

i-0880238b68271661d (My First Server)

IAM Role

IMDSv2
Required

Operator

Subnet ID

subnet-00ce35864b297af51

Instance ARN

arn:aws:ec2:ap-south-2:235562991793:instance/i-0880238b68271661d

Auto Scaling Group name

Managed
false

Instances (1/1) Info

Find Instance by attribute or tag (case-sensitive)

Instance state = running Clear filters

Name	Instance ID	Instance state	Instance type	Status check
My First Server	i-0880238b68271661d	Running	t3.micro	3/3 checks passed

Actions

- Instance diagnostics
- Instance settings
- Networking
- Security
- Image and templates
- Monitor and troubleshoot

Change security groups

Get Windows password

Modify IAM role

Click on modify iam role

i-0880238b68271661d (My First Server)

IAM Role

Subnet ID

subnet-00ce35864b297af51

Instance ARN

arn:aws:ec2:ap-south-2:235562991793:instance/i-0880238b68271661d

Auto Scaling Group name

Managed

false

<https://www.youtube.com/@techwithtwinkle>

EC2 > Instances > i-0880238b68271661d > Modify IAM role

Modify IAM role

Attach an IAM role to your instance.

Instance ID

i-0880238b68271661d (My First Server)

IAM role

Select an IAM role to attach to your instance or create a new role if you haven't created any. The role you select replaces any roles that are currently attached to your instance.

DemoroleEC2

Create new IAM role

No IAM Role

Choose this option to detach an IAM role

DemoroleEC2

arn:aws:iam::235562991793:instance-profile/DemoroleEC2

DemoroleEC2

Cancel

Update IAM role

Select demo role EC2

Click on update iam role

**Go to EC2 type this
your DemouserIAM
Details will be show**

Amazon EC2 Pricing Models

Pricing Model	Description	Best For
On-Demand	Pay per hour or second (no commitment)	Short-term, unpredictable workloads
Reserved Instances (RI)	Commit to 1 or 3 years for a discount	Steady, predictable workloads
Savings Plans	Flexible alternative to RIs, commitment to spend	Consistent workloads across instance families
Spot Instances	Unused capacity at up to 90% discount	Fault-tolerant, batch, or flexible workloads
Dedicated Hosts / Instances	Physical servers just for you	Compliance, licensing, security-sensitive apps
Capacity Reservations	Reserve capacity in a specific AZ	Need guaranteed instance availability

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On-Demand Instances

- ▶ **Pay only for what you use** – per second or per hour.
- ▶ No upfront cost or long-term commitment.
- ▶ **Use Cases:**
 - ▶ Dev/test environments
 - ▶ Short-term apps
 - ▶ Unpredictable workloads
- ▶ **Cons:** Expensive for long-term use.

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Reserved Instances (RI)

- ▶ **Commit** to using EC2 for 1 or 3 years.
- ▶ Get **up to 75% discount** compared to On-Demand.
- ▶ Types of RIs:
 - ▶ **Standard RI** – highest discount, lowest flexibility
 - ▶ **Convertible RI** – can change instance type
 - ▶ **Scheduled RI** – for known time windows
- ▶ **Use Cases:**
 - ▶ Always-on applications
 - ▶ Databases, enterprise workloads
- ▶ **Cons:** Locked into specific instance type/region.

Savings Plans

- ▶ Commit to a **specific amount (\$/hr)** over 1 or 3 years.
- ▶ More flexible than RIs: apply across **instance families, sizes, OS, regions.**
- ▶ Types:
 - ▶ **Compute Savings Plan** – max flexibility (across EC2, Fargate, Lambda)
 - ▶ **EC2 Instance Savings Plan** – cheaper, but locks to family & region
- ▶ **Use Cases:**
 - ▶ Long-running services with varied usage patterns
- ▶ **Cons:** Still a commitment.

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Spot Instances

- ▶ Use **unused EC2 capacity** at up to **90% discount**.
- ▶ **Can be interrupted** with 2-minute notice.
- ▶ **Use Cases:**
 - ▶ Batch processing, ML training
 - ▶ CI/CD jobs
 - ▶ Fault-tolerant applications
- ▶ **Cons:** Not for critical/interactive apps.

Dedicated Hosts / Instances

- ▶ **Dedicated Instances:** Run on isolated hardware (still managed by AWS).
- ▶ **Dedicated Hosts:** You manage physical server (good for software licenses like Windows or Oracle).
- ▶ **Use Cases:**
 - ▶ Bring your own licenses (BYOL)
 - ▶ Regulatory/compliance requirements
- ▶ **Cons:** Higher cost, less elasticity.

Capacity Reservations

- ▶ Reserve capacity in a **specific Availability Zone**.
- ▶ Can combine with Savings Plans or RIs.
- ▶ **Use Cases:**
 - ▶ Mission-critical workloads
 - ▶ Disaster recovery
 - ▶ Event-based apps (e.g. Black Friday scaling)
- ▶ **Cons:** No cost saving unless combined with RIs/Savings Plan.

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Comparison

Model	Cost	Commitment	Interruptible?	Flexibility
On-Demand	High	None	No	Very High
Reserved Instance	Low	1–3 years	No	Low
Savings Plans	Medium-Low	1–3 years	No	Medium-High
Spot	Very Low	None	Yes	High
Dedicated	High	Optional	No	Low
Capacity Reservation	Regular	Optional	No	Medium

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IP Address Charges in AWS

New Public IPv4 Address Pricing (As of Feb 1, 2024)

Amazon now charges \$0.005/hour for all public IPv4 addresses, even if:

Automatically assigned to running EC2

Assigned to services like ALB, NLB, NAT Gateway

That's ~\$3.60/month per public IPv4 address.

Condition	Cost (approx.)
Elastic IP (IPv4) allocated but not attached to a running instance	\$0.005/hour
More than 1 Elastic IP per instance	\$0.005/hour
Public IPv4 address on services like NAT Gateway	Additional data processing cost applies
BYOIP (Bring Your Own IPv4)	Charged for advertising and usage

IPv4 vs IPv6 in AWS – Pricing Overview

Feature	IPv4	IPv6
Address space	Limited	Vast (practically unlimited)
Public IP charge?	✓ Sometimes	✗ Free
Private IP charge?	✗ Free	✗ Free
Elastic IP charge?	✓ Charged if unused	✗ Not applicable (EIP not supported for IPv6)
NAT required for internet?	✓ Yes (for private IPv4)	✗ No (IPv6 has native internet access)
Availability	Default	Optional, must enable

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EC2 Spot Instances – What Are They?

- ▶ **Spot Instances** let you buy unused EC2 capacity at **up to 90% discount** compared to On-Demand prices.
- ▶ **Key Features:**
- ▶ Same performance as On-Demand
- ▶ **Heavily discounted**
- ▶ **Can be interrupted** with a 2-minute warning when AWS needs capacity back
- ▶ Flexible — ideal for fault-tolerant or batch jobs
- ▶ **Example Use Cases:**
- ▶ Machine Learning training
- ▶ CI/CD pipelines
- ▶ Big data processing
- ▶ Video encoding
- ▶ Web crawling

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Spot Instance Pricing:

Model	Price Range	Preemption?
On-Demand	\$\$\$ (Full Price)	✗ No
Spot	\$ (Up to 90% off)	✓ Yes (2-min notice)

Prices vary based on supply and demand in the selected AZ, instance type, and time.

Spot Instance Drawbacks:

May be terminated anytime

No SLA (Service Level Agreement)

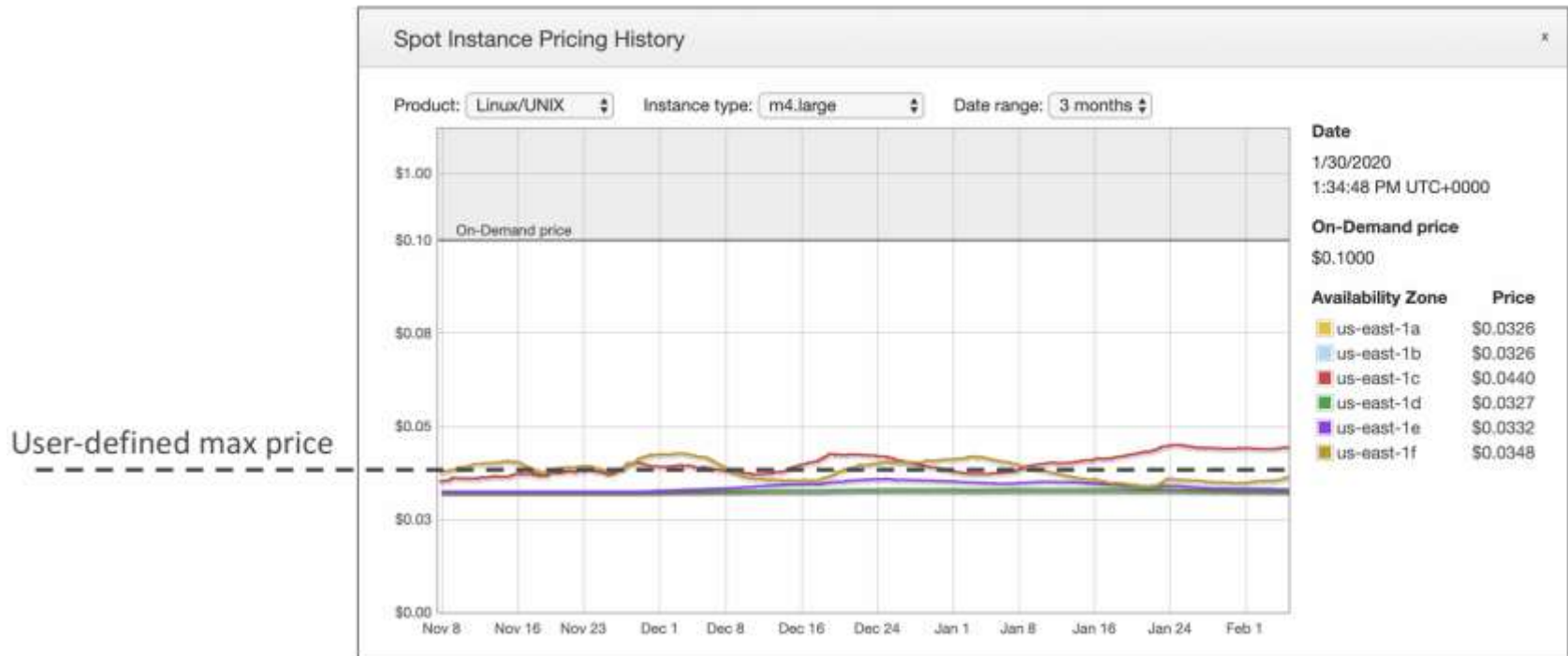
Not good for critical or long-running stateful applications unless fault-tolerant design is in place

Spot block are no longer available to new AWS customers since July 1st 2021 and won't be supported after december 31st 2022

Still they"might" appear in the exam, I won't remove this slide

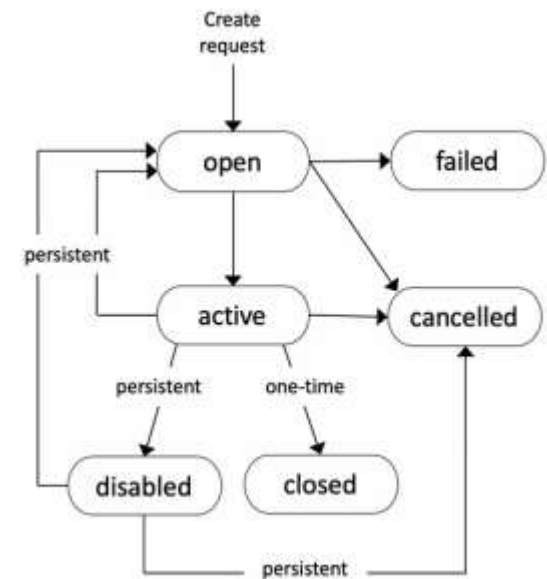
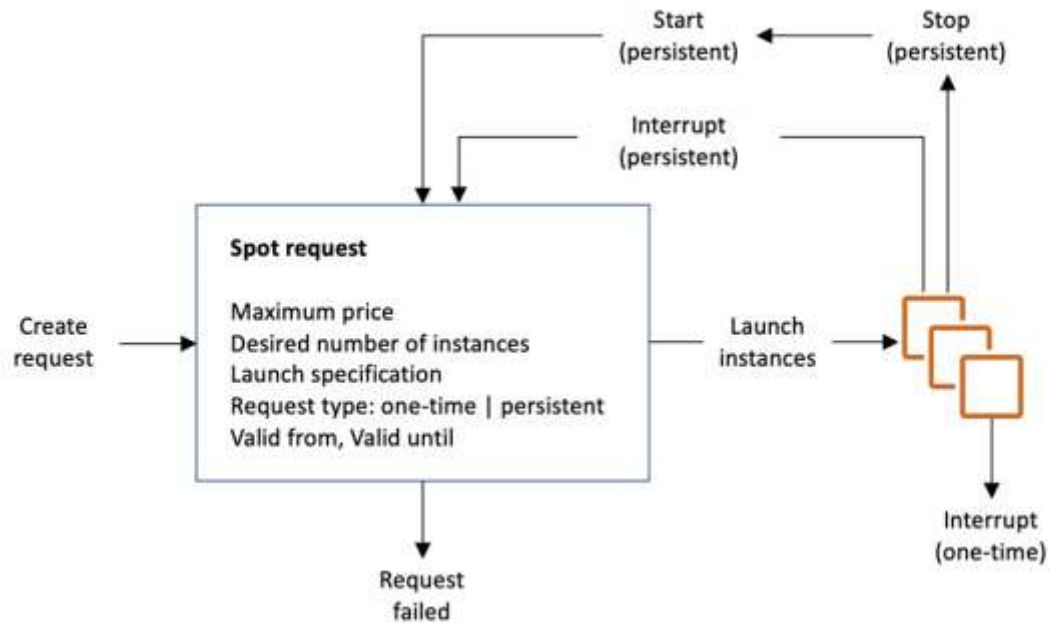
<https://www.youtube.com/@techwithtwinkle>

EC2 Spot Instance Pricing Graph



<https://www.youtube.com/@techwithtwinkle>

How to terminate spot instance?



<https://www.youtube.com/@techwithtwinkle>

Spot Fleet – What Is It?

- ▶ A **Spot Fleet** is a group of Spot Instances launched based on **defined strategy, capacity, and cost controls**.
- ▶ Think of it as a **smart manager** that maintains a desired capacity using Spot (and optionally On-Demand) instances.

Features:

- ▶ Automatically selects **best pricing and instance pools**
- ▶ Supports **diversity across instance types & AZs**
- ▶ Can include **On-Demand fallback**
- ▶ Scales up/down based on target capacity or cost

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Spot Fleet Request Configuration Includes:

Parameter	Example
Target capacity	50 vCPUs or 200 units
Instance types	<code>t3.large</code> , <code>m5.large</code> , <code>c5.xlarge</code>
Allocation strategy	LowestPrice / Diversified / CapacityOptimized
Max price	<code>\$0.03/hour</code>
Pool diversification	Across AZs or types for resilience

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Allocation Strategies in Spot Fleet

Strategy	Description	Best For
LowestPrice	Chooses the lowest-cost Spot pool	Cost-saving priority
Diversified	Spreads across multiple pools	Resilience to interruptions
CapacityOptimized	Chooses pools with low chance of interruption	Long-running workloads
PriceCapacityOptimized	Mixes price and capacity intelligence	Balanced approach (newest, recommended)

<https://www.youtube.com/@techwithtwinkle>

Spot vs Spot Fleet – Quick Comparison

Feature	Spot Instance	Spot Fleet
Scope	Single instance	Group of instances
Manual effort	High	Low (automated)
Diversification	No	Yes (across AZ/types)
Resilience	Low	High
Use cases	One-off jobs, test	Large-scale compute, autoscaling

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EC2 Instances Launch Types Hands On

The screenshot shows the AWS Management Console interface. The top navigation bar includes the AWS logo, a search bar, and various utility icons. The left sidebar contains the navigation menu, with the 'Instances' section expanded and 'Spot Request' highlighted. An orange arrow points to 'Spot Request' with the text 'Click on Spot Request'. The main content area displays the 'Amazon Elastic Compute Cloud (EC2)' title and a 'Launch a virtual server' card with 'Launch instance' and 'View dashboard' buttons.

EC2

- Dashboard
- EC2 Global View
- Events
- ▼ **Instances**
 - Instances
 - Instance Types
 - Launch Templates
 - Spot Request
 - Savings Plans
 - Reserved Instances
 - Dedicated Hosts
 - Capacity Reservations
- ▼ **Images**

Amazon Elastic Compute Cloud (EC2)
Create, manage, and monitor virtual servers in the cloud.

Amazon Elastic Compute Cloud (Amazon EC2) offers the broadest and deepest compute platform with over 600 instance types and a choice of the latest processors, storage, networking, operating systems, and purchase models to help you best match the needs of your workload.

Launch a virtual server

To get started, launch an Amazon EC2 instance, which is a virtual server in the cloud.

Launch instance

View dashboard

<https://www.youtube.com/@techwithtwinkle>

Spot Requests

Search for requests

Request... Request type Instance t... State Capacity Status Persistence Created

You currently have no Spot Fleet requests in this Region.

If you are new to EC2 Spot Instances, visit the [Getting Started page](#).

Choose the Create Spot Fleet request button to launch a fleet of On-Demand and Spot Instances.

Create Spot Fleet request

Spot savings summary

A high-level summary of your savings across all of your running and recently terminated Spot Instances. For detailed reporting on your account-level Spot usage, visit [Cost Explorer](#).

Total Spot Instances
0 instances

Total Spot cost (USD)
\$0.00
You saved 0%

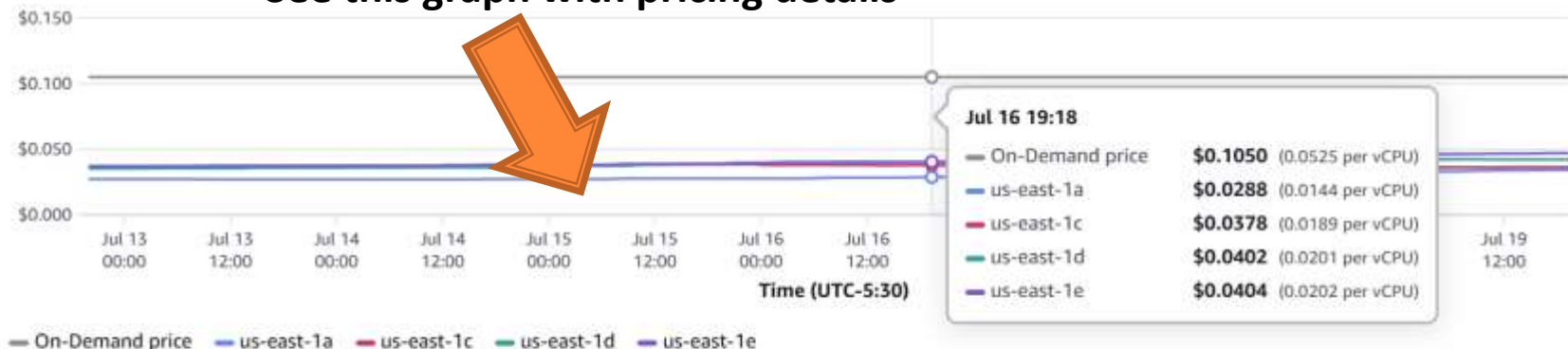
Total usage
0 vCPU hours
0 Memory(GiB) hours

Average cost per usage (USD)
\$0.0000 per vCPU hour
\$0.0000 per Memory hour

You can change instance type, platform & date range to check according to your requirement

Graph: Availability Zones Instance type: c3.large Platform: Linux/UNIX Date range: 1 week

Prices



Average per hour within date range

<https://www.youtube.com/@techwithtwinkle>

aws [ALT+S] United States (N. Virginia) twinkie @ aws-twinkle-v4

EC2 > Spot Requests

Spot Requests

Search for requests

Request... Request type Instance t... State Capacity Status Persistence Created

You currently have no Spot Fleet requests in this Region.

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Choose the Create Spot Fleet request button to launch a fleet of On-Demand and Spot Instances.

[Create Spot Fleet request](#)

Spot savings summary

A high-level summary of your savings across all of your running and recently terminated Spot Instances. For detailed reporting on your account-level Spot usage, visit [Cost Explorer](#).

Total Spot Instances	Total Spot cost (USD)	Total usage	Average cost per usage (USD)
0 instances	\$0.00 You saved 0%	0 vCPU hours 0 Memory(GiB) hours	\$0.0000 per vCPU hour \$0.0000 per Memory hour

Click on Spot Fleet Request

<https://www.youtube.com/@techwithtwinkle>

EC2 > Spot Requests > Create Spot Fleet request

[Create new key pair](#)

► Additional launch parameters -optional

Additional request details

☒ Apply default

IAM fleet role

Keep the request valid from now for 1 year

Terminate the instances when the request expires

Target capacity

Total target capacity

Set your total target capacity (number of instances or vCPUs) to launch. If you specified a launch template, you can allocate part of the target capacity as On-Demand. The number of On-Demand Instances always persists, while Spot Instances can be scaled.

By deselect this you can do according to your requirement

Keep the request valid from now for 1 year
Terminate the instances when the request expires.

Target capacity

Total target capacity

Set your total target capacity (number of instances or vCPU) to launch. If you specified a launch template, you can allocate part of the target capacity as On-Demand. The number of On-Demand instances always persists, while Spot instances can be scaled.

instances

☐ Include On-Demand base capacity

Allocates part of target capacity to On-Demand instances.

☒ Maintain target capacity

Automatically replace interrupted Spot instances.

Interruption behavior

Terminate

☐ Capacity rebalance

When a rebalance notification is sent to a Spot instance, Spot Fleet automatically attempts to replace the instance before it is interrupted. [Learn more](#)

☐ Set maximum cost for Spot instances

Set the maximum amount per hour that you're willing to pay for all the Spot instances in your fleet.

You can change acc to requirement

Network

aws Search [Alt+S] United States (N. Virginia) twinkie @ aws-twinkle-v4

EC2 > Spot Requests > Create Spot Fleet request

► Preview matching instance types (566)

Allocation strategy

Spot allocation strategy

The allocation strategy determines which of your available pools to request Spot instances from.

☒ Price capacity optimized (recommended)

Request the lowest priced Spot instances from your most available pools. This is the best strategy for balancing instance price and interruption risk.

☐ Capacity optimized

Request Spot instances from your most available pools. This strategy has the lowest risk of interruption.

☐ Diversified across all pools

Request Spot instances evenly across all your available pools. This strategy can't be used with attribute-based instance type selection.

Your fleet request at a glance

Total target capacity 10 instances	Instance requirements Attribute based no vCPUs minimum no vCPUs maximum no Mib minimum no Mib maximum	Fleet strength Strong 566 matching instance types 6 Availability Zones	Estimated hourly price ~\$8.002 per hour at target capacity 85% savings compared to On-Demand
--	---	--	--

Cancel [JSON config](#) [Launch](#)

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Click on Launch

Here it will show your fleet request at a glance, if everything ok

Reserved Instance

- ▶ A **commitment-based discount model**. You reserve EC2 instances in a specific AZ, platform, and instance type for **1 or 3 years**.

Key Points:

- ▶ Save **up to 75%** over On-Demand
- ▶ Choose **Standard RI** (more discount) or **Convertible RI** (flexible)
- ▶ **No auto scaling**, unless you plan it

Feature	Value
Discount	Up to 75%
Term	1 or 3 years
Scope	Region or AZ
Payment	All upfront, partial, or no upfront
Instance Flexibility	✗ Only Convertible RIs allow changes

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Reserved Instance

The screenshot displays the AWS Management Console interface for the 'Reserved Instances' page. The top navigation bar includes the AWS logo, a search bar, and the user's account information. The left sidebar shows the 'Instances' section expanded, with 'Reserved Instances' highlighted. The main content area features a blue banner recommending Savings Plans over Reserved Instances. Below this, the 'Reserved Instances' section is visible, showing a message that the user currently has no Reserved Instances in this region. A 'Purchase Reserved Instances' button is prominently displayed. An orange arrow points to this button, and another orange arrow points to the 'Reserved Instances' link in the sidebar.

Click on purchase reserved instances

Click on reserved instance

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Purchase Reserved Instances

☐ Only show offerings that reserve capacity

Platform
Linux/UNIX

Tenancy
Dedicated

Offering class
Convertible

Instance type
t3.micro

Term
1 month to 12 months

Payment option
Partial upfront

Search

By selecting all the details as per your requirements it will show you the pricing

Seller	Term	Effective rate	Upfront price	Hourly rate	Payment option	Offering class	Quantity available	Desired quantity
AWS	12 month	\$0.008	\$34.00	\$0.0039	Partial upfront	Convertible	Unlimited	1

Add to cart

You currently have no items in your cart.

Cancel View cart

Than click on Add to cart

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Purchase Reserved Instances

☐ Only show offerings that reserve capacity

Platform
Linux/UNIX

Tenancy
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Offering class
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Seller	Term	Effective rate	Upfront price	Hourly rate	Payment option	Offering class	Quantity available	Desired quantity
AWS	12 month	\$0.008	\$34.00	\$0.0039	Partial upfront	Convertible	Unlimited	1

Add to cart

Your cart: 1 Reserved Instance, total due now: **\$34.00**
Average monthly recurring cost: **\$2.85**
Additional taxes may apply.

Cancel New cart

It will show you amount

Go to view cart

Purchase Reserved Instances

Save money with an AWS Savings Plan

AWS Savings Plan is an easy-to-use, flexible pricing model that offers discounts on Amazon EC2 instance usage costs. Create a savings plan now to start saving on your Amazon EC2 instance usage costs. [Learn more](#)

You will be billed for every hour during the term of a Partial or No upfront Reserved Instance, regardless of the usage.

Shopping cart

☐	Items to order	Offering class	Region	Upfront price	Hourly price	Order on
☐	Partial upfront Linux/UNIX t3.micro for 12 months Payment terms: Upfront + Hourly (regardless of usage) Sold by: Amazon Edit Delete	Convertible	United States (N. Virginia)	\$34.00	\$0.0039	Now

Your cart: 1 Reserved Instance, total due now: **\$34.00**

Average monthly recurring cost: **\$2.85**

Additional taxes may apply.

Delete selected items (0)

Add more to cart

Order all

When you choose **Order all**, your Reserved Instance is committed and you are immediately charged the upfront fees.

Click on order all if you need them

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Saving Plans

- ▶ The **most flexible discount model**. You commit to a **\$/hour spend** (not specific instances) for 1 or 3 years.

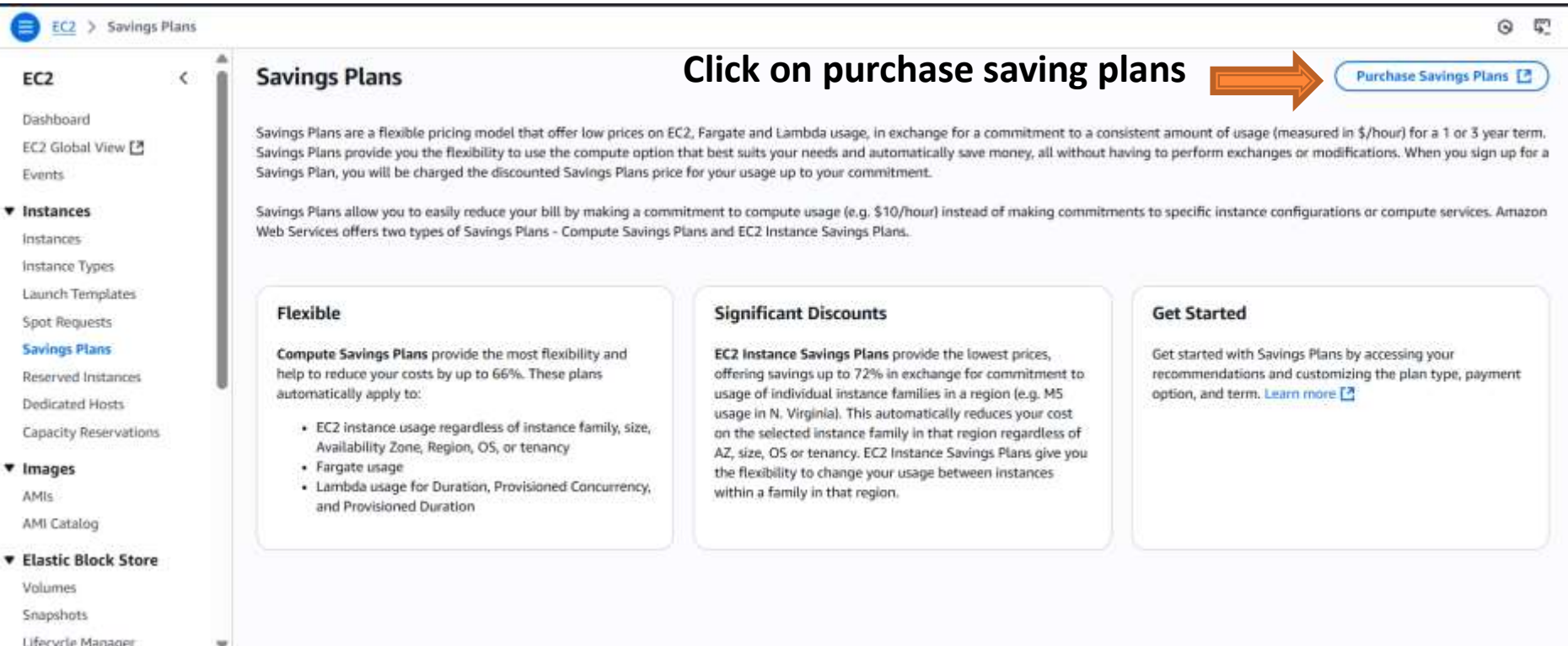
Key Types:

- ▶ **Compute Savings Plan**: Works across any region, instance family, OS, or tenancy
- ▶ **EC2 Instance Savings Plan**: Less flexible but slightly more discount

Feature	Value
Discount	Up to 72%
Term	1 or 3 years
Flexibility	✓ Very high (esp. Compute Plan)
Best for	Autoscaling, modern workloads
Payment Options	All, partial, or no upfront

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Saving Plans



The screenshot shows the AWS Savings Plans console. On the left is a navigation sidebar with links for EC2, Dashboard, EC2 Global View, Events, Instances, Instance Types, Launch Templates, Spot Requests, Savings Plans (highlighted), Reserved Instances, Dedicated Hosts, Capacity Reservations, Images, AMIs, AMI Catalog, Elastic Block Store, Volumes, Snapshots, and Lifecycle Manager. The main content area is titled 'Savings Plans' and includes a description of the pricing model, a 'Purchase Savings Plans' button, and three informational boxes: 'Flexible' (describing Compute Savings Plans), 'Significant Discounts' (describing EC2 Instance Savings Plans), and 'Get Started' (providing instructions on how to begin). An orange arrow points from the text 'Click on purchase saving plans' to the 'Purchase Savings Plans' button.

EC2 > Savings Plans

Savings Plans

Click on purchase saving plans → [Purchase Savings Plans](#)

Savings Plans are a flexible pricing model that offer low prices on EC2, Fargate and Lambda usage, in exchange for a commitment to a consistent amount of usage (measured in \$/hour) for a 1 or 3 year term. Savings Plans provide you the flexibility to use the compute option that best suits your needs and automatically save money, all without having to perform exchanges or modifications. When you sign up for a Savings Plan, you will be charged the discounted Savings Plans price for your usage up to your commitment.

Savings Plans allow you to easily reduce your bill by making a commitment to compute usage (e.g. \$10/hour) instead of making commitments to specific instance configurations or compute services. Amazon Web Services offers two types of Savings Plans - Compute Savings Plans and EC2 Instance Savings Plans.

Flexible

Compute Savings Plans provide the most flexibility and help to reduce your costs by up to 66%. These plans automatically apply to:

- EC2 instance usage regardless of instance family, size, Availability Zone, Region, OS, or tenancy
- Fargate usage
- Lambda usage for Duration, Provisioned Concurrency, and Provisioned Duration

Significant Discounts

EC2 Instance Savings Plans provide the lowest prices, offering savings up to 72% in exchange for commitment to usage of individual instance families in a region (e.g. M5 usage in N. Virginia). This automatically reduces your cost on the selected instance family in that region regardless of AZ, size, OS or tenancy. EC2 Instance Savings Plans give you the flexibility to change your usage between instances within a family in that region.

Get Started

Get started with Savings Plans by accessing your recommendations and customizing the plan type, payment option, and term. [Learn more](#)

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Billing and Cost Management > **Savings Plans** > Overview

Billing and Cost Management

Choose billing view **New**
Choose a billing view

Home
Getting Started
Billing and Payments
Bills
Payments
Credits
Purchase Orders
Cost and Usage Analysis
Cost Explorer
Cost Explorer Saved Reports
Cost Anomaly Detection
Free Tier
Data Exports
Customer Carbon Footprint

Overview Info

Manage alert subscriptions Purchase a Savings Plan

► What are Savings Plans?

Savings Plans summary Info

Active plans (MTD) Savings (MTD) Top savings opportunity Savings Plans expiring this month

Month-to-date

Details

Type	Active plans (MTD)	Commitment (MTD)	Utilization (MTD)	Savings (MTD)
No Active Savings Plans				

Review your savings plan recommendations

Click on purchase Saving plans

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Purchase Savings Plans Info

Savings Plans are a flexible pricing model that offer low prices on AWS usage, in exchange for a commitment to a consistent amount of usage (measured in \$/hour) for a 1- or 3-year term.

Purchase details Info

Savings Plans type

☒ **Compute Savings Plans**
Applies to EC2 instance usage, AWS Fargate, and AWS Lambda service usage, regardless of region, instance family, size, tenancy, and operating system.
[Learn more](#)

☐ **EC2 Instance Savings Plans**
Applies to instance usage within the committed EC2 family and region, regardless of size, tenancy, and operating system.
[Learn more](#)

☐ **SageMaker Savings Plans**
Applies to SageMaker service usage, regardless of Region, instance family, and component.
[Learn more](#)

Term
The commitment term for your Savings Plans.
Select a commitment term

Purchase commitment Info

Hourly commitment
Your hourly commitment at Savings Plan rates. See applicable rates for Savings Plans [here](#). To maximize your savings, see our [recommendation](#).
Enter hourly commitment amount (\$)

You can choose according to requirements & add to cart complete the purchasing

Dedicated Host

- ▶ A **physical server fully dedicated** to your use. You get **complete control** over instance placement, sockets, cores, etc.

Best For:

- ▶ Software that needs **per-socket or per-core licensing** (Oracle, SQL Server)
- ▶ Compliance requirements (hardware isolation)
- ▶ BYOL (Bring Your Own License)

Feature	Value
Pricing	\$\$\$ (costly)
Commitment	Optional
Visibility	Full hardware access
Billing	Per host, not per instance

Dedicated Host

The screenshot displays the AWS Management Console interface for the 'Dedicated Hosts' section. The left-hand navigation pane includes links to 'Dashboard', 'EC2 Global View', 'Events', 'Instances', 'Images', and 'Elastic Block Store'. The main content area, titled 'Dedicated Hosts', features a search bar and a table with columns for Name, Host ID, Availability, State, vCPU utilization, Instance, and Host Reservation. A message states 'No Dedicated Hosts. You do not have any Dedicated Hosts in this region.' Below this message is a button labeled 'Allocate Dedicated Host'. An orange arrow points from the text 'Click on Allocate Dedicated Host' to the 'Allocate Dedicated Host' button in the top right corner of the main content area.

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Allocate Dedicated Host

Dedicated Hosts allow you to provision EC2 instances on physical servers fully dedicated for your use.

Dedicated Host settings

Name tag

test

Instance family [Info](#)

Specifies the instance family to be supported by the Dedicated Host.

c5

Support multiple instance types [Info](#)

Launch different instance types from the same instance family on the Dedicated Host.

☒ Enable

Availability Zone [Info](#)

The Availability Zone in which the Dedicated Host is located.

use1-az6 (us-east-1a)

Outpost ARN - optional

arn:aws:outposts:[region]:[account]:outpost/[outpost-id]

Instance auto-placement [Info](#)

Disabling auto-

☐ Enable

Enter tag name

Select instance family

Select Availability Zones

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Select As per your requirements Than click on Allocate

Host recovery [Info](#)

Automatically restarts your instances on to a new replacement host if your current host is unreachable due to a permanent failure.

☐ Enable

Host maintenance [Info](#)

Automatically allocates a new dedicated host when degradation is detected on current host, and reboots your instances on the new host during a scheduled maintenance event. Degraded hosts are reachable until a maintenance event is initiated.

☒ Enable

Quantity

The number of Dedicated Hosts you'd like to purchase for this configuration.

Note: You will be billed per allocated Dedicated host, not per instance. [Learn more](#)

1

▶ CLI commands

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Q Name

Value - optional

Q test

Remove

Add new tag

You can add 49 more tags

Cancel

Allocate

Capacity Reservations

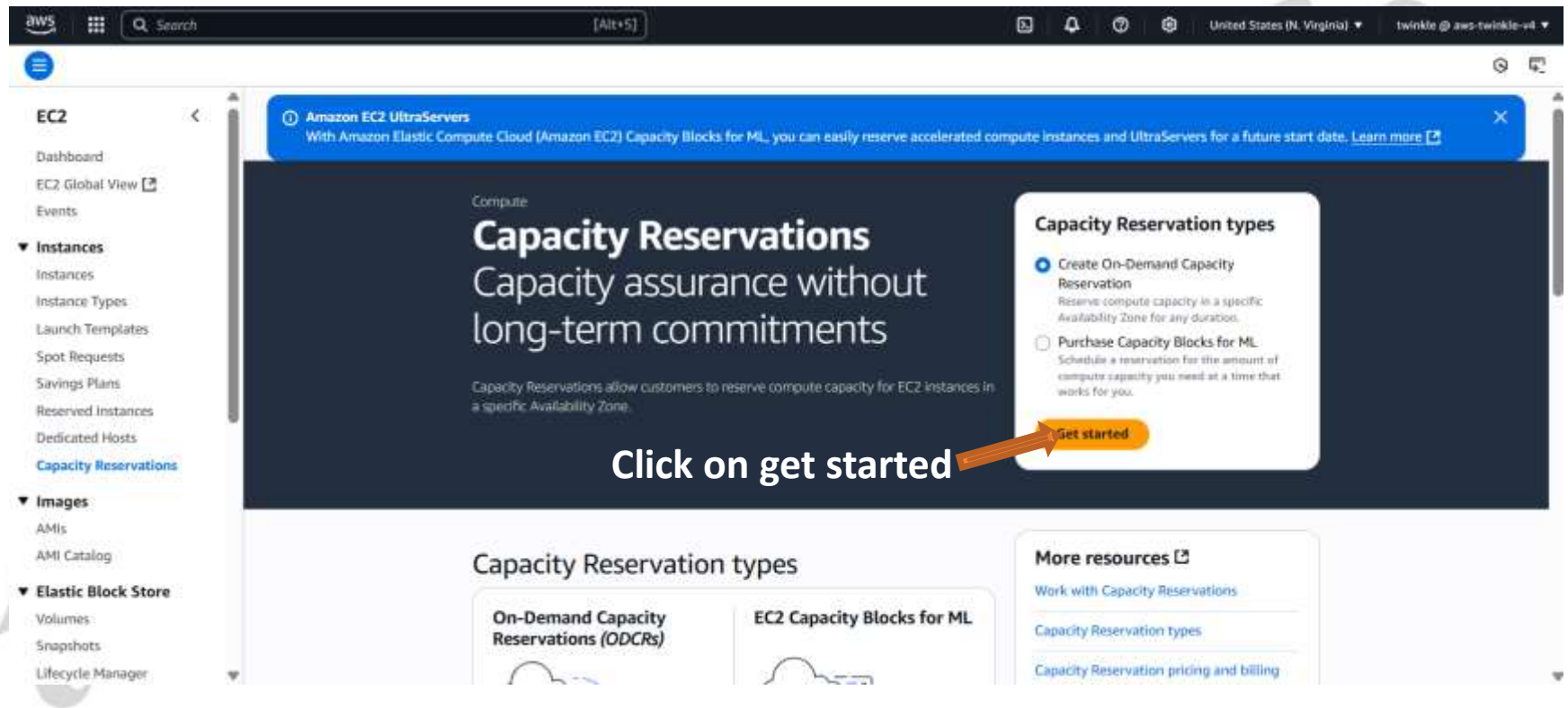
- ▶ Guarantees EC2 instance capacity in a specific AZ, so your instance **always launches**, even during high demand.

Key Points:

- ▶ You **pay On-Demand price**
- ▶ Can be **targeted** (only used by specific instances) or **untargeted**
- ▶ Good for **HA workloads** or **regulatory needs**
- ▶ No long-term commitment (can cancel anytime)

Feature	Value
Pricing	On-Demand
Commitment	None
Use Case	Capacity guarantee
AZ Specific?	☒ Yes

Capacity Reservations



Amazon EC2 UltraServers
With Amazon Elastic Compute Cloud (Amazon EC2) Capacity Blocks for ML, you can easily reserve accelerated compute instances and UltraServers for a future start date. [Learn more](#)

Capacity Reservations

Capacity assurance without long-term commitments

Capacity Reservations allow customers to reserve compute capacity for EC2 instances in a specific Availability Zone.

Capacity Reservation types

- ☒ **Create On-Demand Capacity Reservation**
Reserve compute capacity in a specific Availability Zone for any duration.
- ☐ **Purchase Capacity Blocks for ML**
Schedule a reservation for the amount of compute capacity you need at a time that works for you.

[Get started](#)

Click on get started

Capacity Reservation types

- On-Demand Capacity Reservations (ODCRs)**
- EC2 Capacity Blocks for ML**

More resources

- [Work with Capacity Reservations](#)
- [Capacity Reservation types](#)
- [Capacity Reservation pricing and billing](#)

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EC2 > Capacity Reservations > Create On-Demand Capacity Reservation

EC2

- Dashboard
- EC2 Global View
- Events
- Instances
 - Instances
 - Instance Types
 - Launch Templates
 - Spot Requests
 - Savings Plans
 - Reserved Instances
 - Dedicated Hosts
 - Capacity Reservations
- Images
 - AMIs
 - AMI Catalog
- Elastic Block Store
 - Volumes
 - Snapshots
 - Lifecycle Manager

Create On-Demand Capacity Reservation Info

Create a Capacity Reservation to reserve compute capacity for your instances in a specific Availability Zone for any duration.

Instance details Info

Instance type
a1.medium

☒ EBS-optimized

Platform
Linux/UNIX

Availability Zone
us-east-1a

Tenancy
Default - run a shared hardware instance

Placement group
None

Total capacity
You can reserve capacity up to your current On-Demand instance quota.

Choose according to requirements click on create

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EC2 > Capacity Reservations > Create On-Demand Capacity Reservation

Specify when the Capacity Reservation should end:

☒ **At a specific time**
The Capacity Reservation will end automatically at a date and time that you specify.

☐ **Manually**
The Capacity Reservation remains active in your account until you manually cancel it.

End date (UTC+05:30)
YYYY/MM/DD hh:mm:ss

Note: By default, the Capacity Reservation is configured to end at a specific date and time that you must specify. This ensures that the reservation is released and billing ends when you expect. If you prefer to keep the reservation indefinitely and cancel it when you no longer need it, select Manually.

Instance eligibility Info
Indicate the criteria for instances that can run in this Capacity Reservation.

☒ **Open: Any instance with matching attributes.**
All instances with matching attributes (instance type, platform, and Availability Zone) automatically run in the capacity.

☐ **Targeted: Only instances that target the Capacity Reservation.**
When you launch instances, you must specify the reservation ID or the reservation resource group ARN associated with this reservation.

Tags (0)
No tags associated with the resource.

[Add new tag](#)

You can add up to 50 more items.

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[Cancel](#) [Create](#)

Summary of Instances

Feature	Capacity Reservation	Dedicated Host	Reserved Instances	Savings Plans
Guarantees capacity	✓ Yes	✓ Yes	✓ Yes	✗ No
Pricing model	On-Demand	Per host	Discounted	Discounted
Commitment	None	Optional or 1/3 yrs	1 or 3 years	1 or 3 years
Best for	HA workloads	Licensing compliance	Predictable workloads	Flexible scaling
Discount?	✗ No	✗ (unless used with RIs/SPs)	✓ Yes	✓ Yes
Instance Flexibility	✗	✗	✗ (Only with Convertible)	✓ (high)

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Which Should You Choose?

Situation	Best Option
Want guaranteed capacity for short time	Capacity Reservation
BYOL licensing, physical separation	Dedicated Host
Steady, predictable workloads	Reserved Instances
Auto-scaling, containerized workloads	Savings Plans (Compute)
Licensing with visibility into sockets/cores	Dedicated Host + RIs

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